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Mid-Mesoproterozoic bimodal magmatic rocks in the northern North China Craton: Implications for magmatism related to breakup of the Columbia supercontinent

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ABSTRACT

Large volumes of diabase sills occur within the late Paleoproterozoic to Mesoproterozoic sedimentary rocks in the northern part of the North China Craton (NCC) and previous geochronological studies suggest some of these were emplaced in the Mid-Mesoproterozoic (Ectasian). Here we present new zircon and baddeleyite SHRIMP, LA-ICP-MS and SIMS U–Pb analyses on representative samples of the diabase sills emplaced into the Wumishan, Tieling and Xiamaling Formations. The results show emplacement ages ranging from 1325 ± 5 Ma to 1316 ± 37 Ma, suggesting voluminous magmatism leading to the formation of the diabase sill swarms (the Yanliao mafic sill swarms) in the Mid-Mesoproterozoic. Zircon LA-ICP-MS U–Pb dating of the Shangdu-Huade granitic pluton in the northern NCC yielded similar emplacement ages ranging from 1331 ± 11 Ma to 1313 ± 17 Ma. These ages are comparable with the 1.4–1.2 Ga rift-related anorogenic magmatism recorded elsewhere on the globe such as North America, Greenland, Baltica, and Siberia, associated with the breakup of the Columbia (Nuna) supercontinent. The Mid-Mesoproterozoic magmatic rocks in the northern NCC constitute a typical bimodal magmatic suite. The geochemical and Nd–Hf isotopic data suggest that the diabasic rocks were likely generated by partial melting of the depleted asthenosphere mantle coupled with slight crustal assimilation in a continental rift setting; however, the granitic rocks were generated mainly through partial melting of the ancient continental crust, probably induced by the upwelling of hot asthenosphere mantle during continent rifting processes. The recognition of these bimodal magmatic rocks indicates that the northern NCC experienced Mid-Mesoproterozoic rifting events that are considered to have led to the final breakup of the Columbia supercontinent. The breakup of the NCC from the Columbia supercontinent occurred during Mid-Mesoproterozoic time, probably at around 1.33 Ga. The carbonatite rocks and rare earth element-niobium mineralization in the giant Bayan Obo ore deposit in the northern NCC were also probably related to the continental rifting event in the Mid-Mesoproterozoic.

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1. Introduction

The assembly, configuration, evolution and breakup of the supercontinents during Earth's history such as Columbia (Nuna), Rodinia, Gondwana and Pangea have been topics of wide interest in recent years (e.g., Hoffman, 1991; Moores, 1991; Dalziel, 1991, 1997; Windley, 1995; Karlstrom et al., 2001; Condie, 2002; Meert, 2002; Meert and Lieberman, 2008; Rogers and Santosh, 2002, 2003, 2009; Zhao et al., 2002a,b,c, 2003a,b, 2004, 2006; Pesonen et al., 2003; Kröner and Cordani, 2003; Ernst et al., 2008; Li et al., 2008; Murphy et al., 2009a,b; Reddy and Evans, 2009; Evans, 2009; Kusky and Santosh, 2009; Santosh et al., 2009; Santosh, 2010a; Wang et al., 2010a; Weber et al., 2010; Audet and Bürgmann, 2011). The processes attending the amalgamation and dispersal of supercontinents are considered to have significantly impacted the mantle dynamics, solid earth processes, surface environments, metallogeny and biogeochemical cycles (e.g., Worsley and Nance, 1989; Condie, 2001; Mao and Zhong, 2001; Teixeira et al., 2007; Santosh, 2010b,c; Sawaki et al., 2010). The Paleo-Mesoproterozoic (2.1–1.2 Ga) is regarded as the time of assembly, growth and breakup of the Columbia supercontinent (Zhao et al., 2002a, 2003a, 2004; Ernst et al., 2008; Goldberg, 2010). The assembly of Columbia was probably completed during the global-scale 2.0–1.8 Ga collisional event (e.g., Zhao et al., 2002a, 2003a, 2004). Its disruption commenced at about 1.6 Ga with the final breakup during the period of 1.3–1.2 Ga, as indicated by widespread Mesoproterozoic continental rifting, anorogenic magmatism and emplacement of

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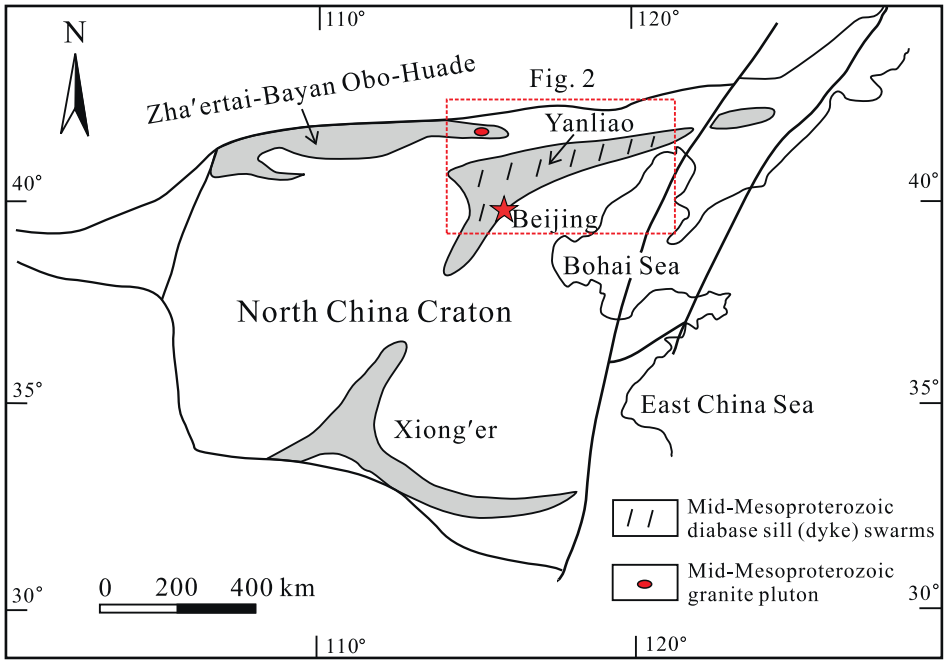


Fig. 1. Sketch map showing late Paleoproterozoic–Mesoproterozoic continental rifting in the NCC (modified after Zhai and Liu, 2003; Zhai, 2004).

mafic dyke swarms in all the cratonic blocks which were part of the Columbia supercontinent (e.g., Zhao et al., 2003a, 2004, 2006; Hou et al., 2008; Goldberg, 2010).

The assembly of the Columbia supercontinent has been recorded in the North China Craton (NCC) by the 2.0–1.8 Ga collisional orogenic belts including the Trans-North China Orogen (Zhao et al., 2003a, 2004; Zhang et al., 2009a; Zhao, 2009; Li et al., 2010a) and the Khondalite Belt (Zhao et al., 2003c, 2005; Xia et al., 2008; Yin et al., 2009, 2011) or the Inner Mongolia Suture Zone (Santosh, 2010d). However, whether the NCC was involved in the breakup of the Paleo-Mesoproterozoic supercontinents is still controversial (e.g., Zhai et al., 2000; Lu et al., 2008; Rogers and Santosh, 2002; Zhao et al., 2003a; Li et al., 2008), mainly due to lack of reliable isotopic ages for the Mesoproterozoic (1.6–1.0 Ga) magmatism and tectonism in the NCC. Some researchers believe that the continental rifting in the northern NCC ended prior to 1.6 Ga (e.g., Rogers and Santosh, 2002; Zhai, 2004; Lu et al., 2008), and that the NCC was not involved in the final breakup of the Columbia supercontinent during 1.35–1.20 Ga (e.g., Rogers and Santosh, 2002; Hou et al., 2008) nor the assembly and breakup of the Rodinia supercontinent in the late Mesoproterozoic to Neoproterozoic (e.g., Lu et al., 2008). However, a comparison of the available paleomagnetic data from the NCC (Halls et al., 2000; Wu et al., 2005; Pei et al., 2006) with those from Laurentia (Emslie et al., 1976; Irving et al., 1977; Harlan et al., 1994; Harlan and Geissman, 1998; Halls and Heaman, 2000; Meert and Stuckey, 2002), Siberia (Ernst et al., 2000) or India (Piper et al., 2011) shows that the NCC shares similar paleomagmatic poles with Laurentia, Siberia or India during late Paleoproterozoic to early Mesoproterozoic time, indicating these continental fragments drifted together or close during this period. Paleomagnetic data from the late Mesoproterozoic sedimentary rocks (Yunmengshan Formation, Rb–Sr age ca. 1270 Ma) display a slightly different configuration involving a ca. 90° vertical-axis rotation of the NCC with respect to Laurentia during 1.35–1.20 Ga (Zhang et al., 2006a; Li et al., 2008), indicating that breakup of the NCC from the Columbia supercontinent might have occurred during 1.35–1.20 Ga.

The processes accompanying the breakup of Columbia in the Mid-Mesoproterozoic (Ectasian) in the NCC should be manifest by continental rifting and magmatism comparable with those in other cratons that were involved in the final breakup of this supercontinent. In this contribution, we present systematic field, petrological, geochronological, geochemical and Nd–Hf isotopic studies on the recently recognized Mid-Mesoproterozoic diabase sill swarms and granites in the northern NCC. These bimodal magmatic rocks provide important evidence for the breakup of the NCC from the Columbia supercontinent in the Mid-Mesoproterozoic at around 1.33–1.31 Ga. The results presented in this study provide critical constraints on the processes and timing of disruption of the Columbia supercontinent.

2. Geological setting and previous studies

The NCC is one of the oldest cratons in the world, with crustal rocks as old as 3.8 Ga (Liu et al., 1992; Song et al., 1996; Wan et al., 2005; Wu et al., 2008). After its final cratonization in the Paleoproterozoic at around 1.85–1.80 Ga (e.g., Zhao et al., 2001, 2008a,b, 2010; Zhai, 2004; Kröner et al., 2006; Liu et al., 2006; Zhang et al., 2006b, 2007a; Wang et al., 2010b), the NCC underwent late Paleoproterozoic extension with emplacement of mafic dyke swarms and an anorthosite–mangerite–alkali granitoid–rapakivi granite suite in its northern and central parts during 1.78–1.68 Ga (e.g., Halls et al., 2000; Wang et al., 2004; Yang et al., 2005; Peng et al., 2005, 2008; Zhang et al., 2007b; Zhao et al., 2009a). At least two continental rift zones termed the Yanliao and Zha'ertai-Bayan Obo-Huade rifts developed in the northern NCC (Fig. 1), with a thick sequence of late Paleoproterozoic to Meso-Neoproterozoic sedimentary rocks (Changcheng, Jixian and Qingbaikou Systems) that unconformably covered the Archean–Paleoproterozoic basement metamorphic rocks (Fig. 2). Some K-rich volcanic rocks were indentified from the late Paleoproterozoic Dahongyu Formation with zircon U–Pb ages ranging from 1622 ± 23 Ma to 1626 ± 9 Ma (Lu and Li, 1991; Lu et al., 2008; Gao et al., 2008a), and were considered to be related to continental rift process in the late Paleoproterozoic associated with the dispersal of the supercontinent Columbia (Lu et al., 2008). Recent studies identified the presence of thin K-bentonite beds in the middle segment of the Xiamaling Formation (Gao et al., 2007, 2008b; Su et al., 2008) and the upper

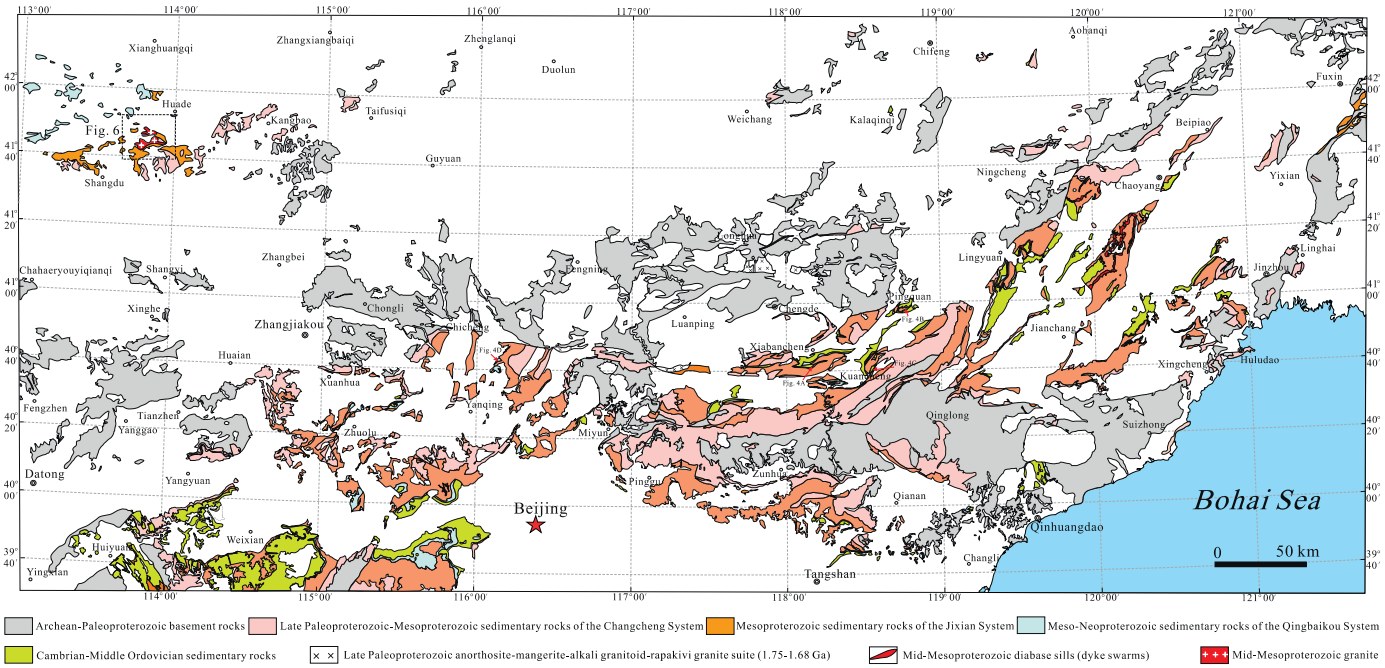


Fig. 2. Sketch geological map of the northern NCC showing distribution of the Archean–Paleoproterozoic basement rocks, the late Paleoproterozoic intrusive rocks, Mid-Mesoproterozoic granites and late Paleoproterozoic to Meso-Neoproterozoic and Early Paleozoic sedimentary rocks in the northern domain (modified after Zhang et al., 2009b). Locations of cross sections in Fig. 4 are indicated. Open square locates region of detailed map in Fig. 6.

segment of the Tieling Formation (Su et al., 2010). Among these, the K-bentonite beds from the Xiamaling Formation yielded mean $^{207}Pb/^{206}Pb$ ages of 1366 ± 9 Ma to 1380 ± 36 Ma (Gao et al., 2007, 2008b; Su et al., 2008, 2010), and those from the Tieling Formation show mean $^{207}Pb/^{206}Pb$ ages of 1437 ± 21 Ma (Su et al., 2010). These new age data provide important constraints on the Mesoproterozoic strata in the NCC (Fig. 3). The K-bentonite beds within the Xiamaling and Tieling Formations in northern NCC are considered to be altered from felsic-intermediate tuffs generated through explosive volcanism (Gao et al., 2007, 2008b; Su et al., 2008, 2010). However, because thin tuff beds can be transported for hundreds of kilometers in the air before deposition, whether their source areas are in the NCC or other nearby cratons is still uncertain (Lu et al., 2010).

Swarms of diabase sills are widespread within the late Paleoproterozoic to Mesoproterozoic marine clastic and carbonate platformal sedimentary rocks in the northern NCC and extend for several hundred kilometers from west to east. Because of the difficuties in dating mafic rocks, these diabase sills were variously regarded as Meso-Neoproterozoic (HBGMR, 1972), Late Paleozoic (Li and Li, 2005), Triassic (LBGMR, 1965, 1967; HBGMR, 1969) or even Jurassic (LBGMR, 1969) in age, mainly based on imprecise whole-rock K–Ar ages. Recent LA-ICP-MS zircon and baddeleyite U–Pb dating on a sample from the diabase sill emplaced into the Mesoproterozoic Wumishan Formation yielded weighed mean $^{207}Pb/^{206}Pb$ ages of 1345 ± 12 Ma (95% confidence, MSWD = 1.8, N = 18) and 1353 ± 14 Ma (95% confidence, MSWD = 0.67, N = 15), respectively (Zhang et al., 2009b). TIMS U–Pb dating on four baddeleyite grains from a diabase sill emplaced into the Xiamaling Formation in Kuancheng yielded weighed mean $^{206}Pb/^{238}U$ age of 1320 ± 6 Ma (95% confidence, MSWD = 1.04) and $^{207}Pb/^{206}Pb$ age of 1325 ± 31 Ma (95% confidence, MSWD = 0.0024), respectively (Li et al., 2009a). The above results indicate that the emplacement of these large volumes of diabase sills occurred in the Mid-Mesoproterozoic and the data suggest a very important tectonomagmatic event in the NCC at this time, probably related to breakup of the Columbia supercontinent (Zhang et al., 2009b).

Although some Mesoproterozoic granitoids were previously dated in the northern NCC based on conventional single-grain zircon U–Pb ages (e.g., IMBGMR, 1997, 1999; HBGMR, 2003), recent zircon SHRIMP U–Pb investigations indicate they are either Archean–Paleoproterozoic or Late Paleozoic in age (Zhang et al., 2007c, 2009c). Thus, the existence of Mesoproterozoic granitoids and related felsic igneous rocks has not so far been conclusively demonstrated from precise geochronological investigations in the NCC. In this contribution, we report for the first time age data on a Mid-Mesoproterozoic granite pluton from the Zha'ertai-Bayan Obo-Huade rift in the northern NCC.

3. Field occurrence, petrology and sample description

3.1. Field occurrence and petrology of diabase sill swarms

Diabase sills are widespread within the late Paleoproterozoic to Mesoproterozoic sedimentary rocks in the northern NCC and are especially common near Chaoyang, Lingyuan, Pingquan, Xiabancheng, Kuancheng, Huailai and western Beijing (Fig. 2). These sills have thickness of several meters up to several hundard meters (Fig. 5A–H). Their intrusive contacts are very clear and thin baked zones or thermal recrystallization can be observed near the boundary of the sills. These diabase sills are particularly common within the Xiamaling and Wumishan Formations; also locally within the Chuanlinggou, Gaoyuzhuang, Hongshuizhuang and Tieling Formations (Fig. 3). Usually, three to five layers of diabse sills can be observed within the Xiamaling Formation from western Bejing to Chaoyang in the northern NCC (Fig. 4). Large volumes of diabase sills are very common within the Wumishan Formation near Chaoyang and are several tens of meters to several hundred meters thick, many of them running to few kilometers or even up to several tens of kilometers (Fig. 5D).

All the sills emplaced within the late Paleoproterozoic to Mesoproterozoic sedimentary rocks display typical diabasic texture (Fig. 5L) with similar mineral compositions of pyroxene (35–60 vol.%), plagioclase (40–55 vol.%), magnetite (3–10 vol.%) and

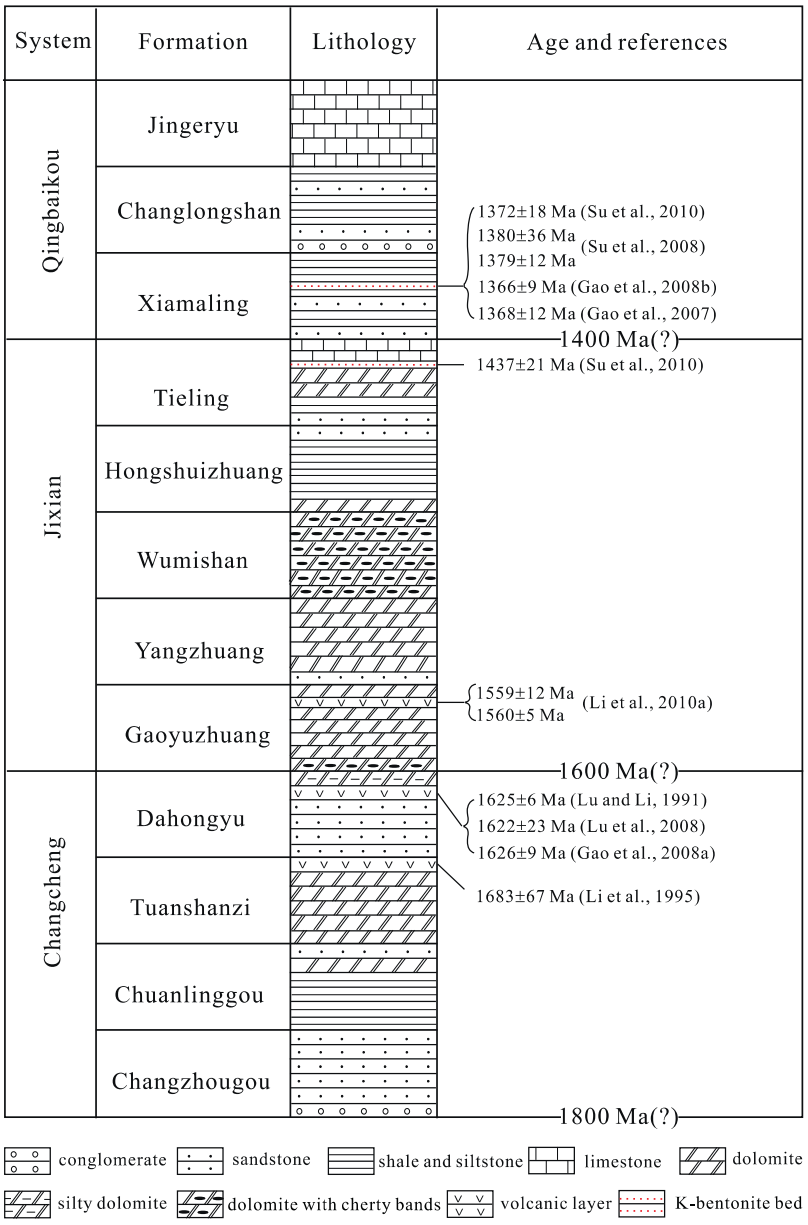


Fig. 3. Subdivision and zircon U–Pb age constraints on late Paleoproterozoic to Meso-Neoproterozoic strata in the northern NCC (modified after Zhang et al., 2009b).

hornblende (0–5 vol.%). Baddeleyites and zircons are common in some of the diabase rocks, especially in the coarse-medium grained varieties.

3.2. Field occurrence and petrology of granitic pluton

The Mid-Mesoproterozoic granitic pluton identified in this study is located between the Shangdu and Huade counties in the northern margin of the NCC (Figs. 2 and 6). It is emplaced into the Mesoproterozoic Jixian System of Bayan Obo Group composed mainly of sedimentary rocks subjected to low-grade metamorphism in the east edge of the Zha'ertai-Bayan Obo-Huade rift zone (Fig. 1). The pluton is intruded by Late Permian and Jurassic–Cretaceous granitoids (Fig. 6), and was previously regarded as Late Paleozoic in age (IMBGMR, 1972), although no precise isotopic dates are available.

Rocks from the granite pluton display coarse-medium grained porphyritic texture with a broadly uniform mineral composition of K-feldspar (40–60 vol.%), plagioclase (5–30 vol.%), quartz (25–35 vol.%), biotite (5–10 vol.%), hornblende (0–5 vol.%) and accessory magnetite, titanite, apatite and zircon, and secondary (<5 vol.%) sericite, muscovite and epidote (Fig. 5I–K). The phenocryst phases are mainly composed of K-feldspar crystals ranging in length from 30 mm to 5 mm. In most places, the rocks are strongly deformed displaying augen structures (Fig. 5I) with quartz recrystallization and undulose extinction (Fig. 5K).

3.3. Sample description

Sample (08093-1) was collected from a diabase sill emplaced into the Wumishan dolomitic limestone near a road cut (GPS position: E120°18.404′; N41°24.709′) southwest of Chaoyang. It is composed of pyroxene (40 vol.%), plagioclase (55 vol.%) and magnetite (5 vol.%), with accessory apatite, zircon and baddeleyite. Zircons from this sample are semitransparent, creamy white subhedral long or short prisms with a size ranging from 60 μm to 150 μm and length/width ratio of 1:1–3.5:1 (Fig. 7A). Baddeleyites are semitransparent or opaque, brown euhedral platy prisms with a size

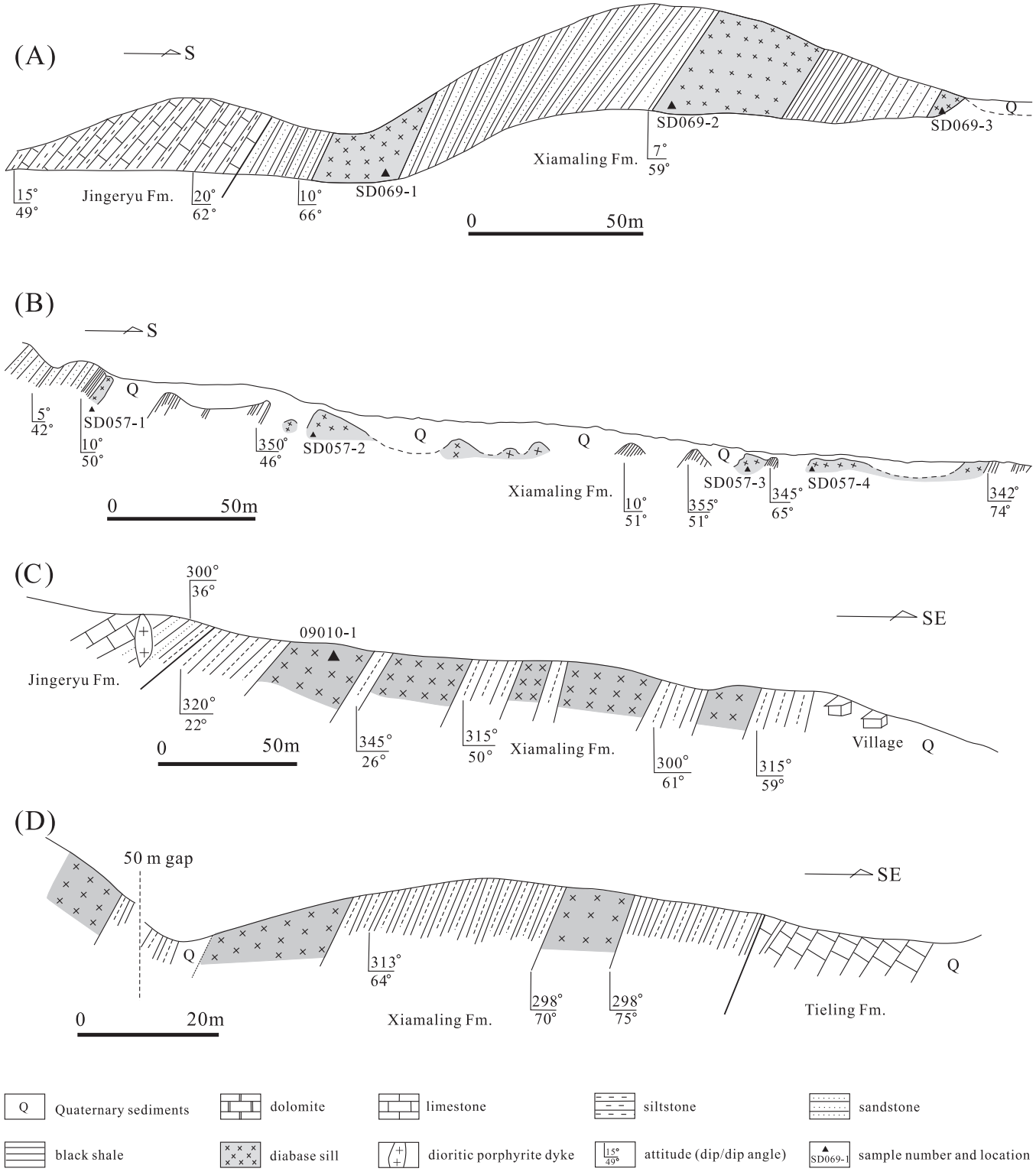


Fig. 4. Representative geological sections showing the diabase sills within the Mesoproterozoic Xiamaling Formation. (A) Wulongji section in Xiabancheng (modified after Zhang et al., 2009b); (B) Shuangdongzi section in Pingquan (modified after Zhang et al., 2009b); (C) Longxumen section in Kuancheng; (D) Zhengjiayao section in Huailai. Locations of sections have been marked in Fig. 2.

ranging from 50 μm to 150 μm and length/width ratio of 1:1–6.5:1 (Fig. 7B).

Sample 09053-1 is another diabase sample collected from a sill within the Wumishan dolomite southwest to Chaoyang (GPS position: E120°18.739′; N41°23.749′). The sample consists of pyroxene (55 vol.%), plagioclase (40 vol.%), hornblende (3 vol.%) and magnetite (2 vol.%), with accessory apatite, zircon and baddeleyite. Zircons from this sample are creamy white subhedral long prisms with a size ranging from 50 μm to 150 μm (Fig. 7C).

Sample 08100-1 was collected from a diabase sill emplaced into the Tieling Formation southwest to Chaoyang (GPS position: E120°14.193′; N41°14.130′). It is composed of plagioclase

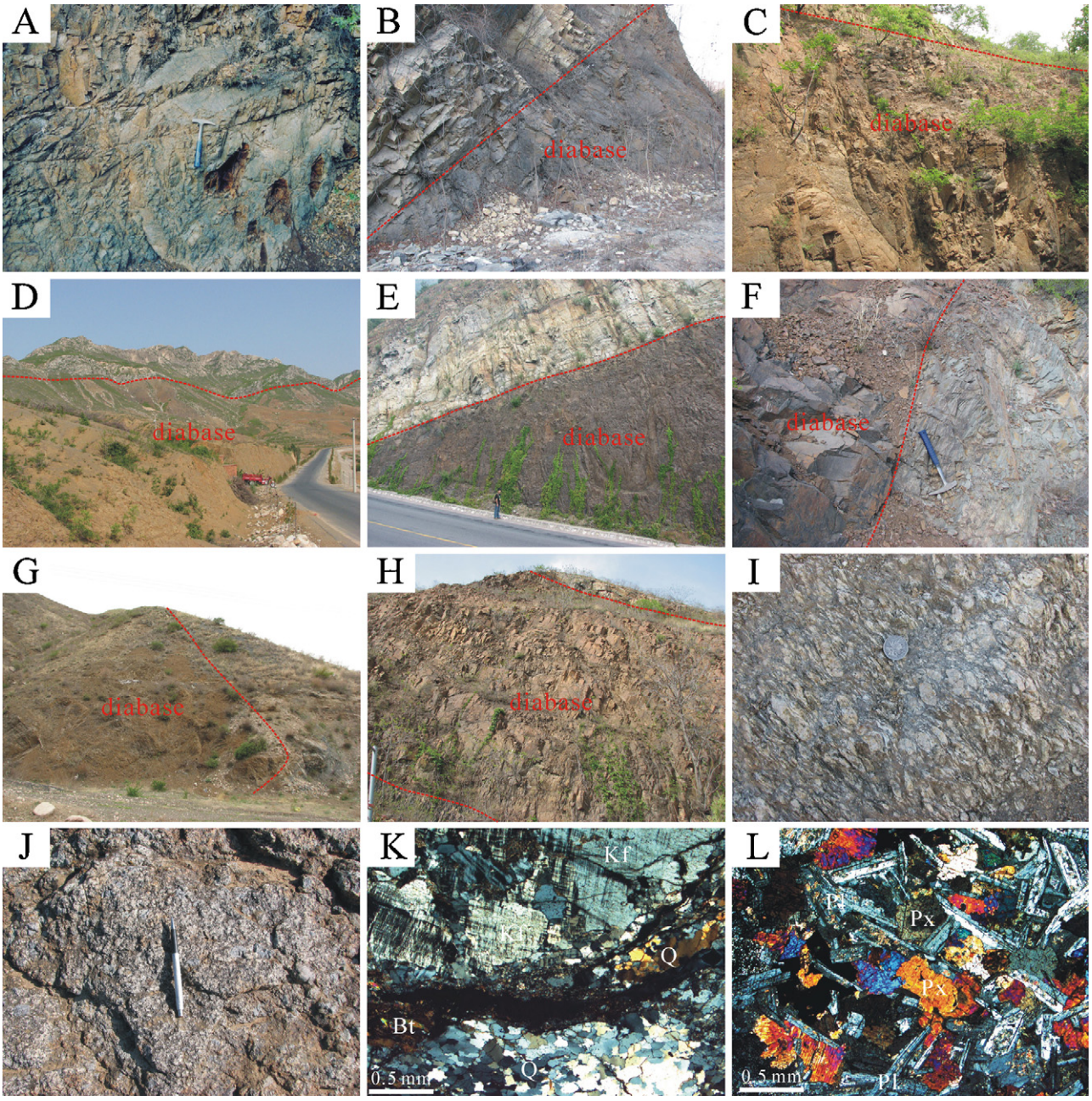


Fig. 5. Representative outcrop photos (A–J) and photomicrographs (K–L) of the Mid-Mesoproterozoic magmatic rocks in the northern NCC. (A) Diabase sill emplaced into the Xiamaling Formation in Shuangdongzi, Pingquan. (B) Diabase sill emplaced into the Xiamaling Formation in Wulongji, Xiabancheng. (C) Diabase sill emplaced into the Xiamaling Formation in Dayingzi, Lingyuan. (D) Diabase sill emplaced into the Wumishan Formation in Chaoyang. (E) Diabase sill emplaced into the Xiamaling Formation in Kuancheng. (F) Diabase sill emplaced into the Xiamaling Formation in Zhengjiayao, Huailai. (G) Diabase sill emplaced into the Xiamaling Formation in Xinbaoan, Huailai. (H) Diabase sill emplaced into the Xiamaling Formation in Qingbaikou, western Beijing. (I) Strongly deformed granite exhibiting augen structures in the Shangdu-Huade pluton. (J) Less deformed granite with coarse grained porphyritic texture in the Shangdu-Huade pluton. (K) Typical mineral assemblage in Mid-Mesoproterozoic granite (sample 07014-2). (L) Representative photomicrograph of Mid-Mesoproterozoic diabase sill (sample 09038-1 from Lingyuan). Mineral abbreviations in photomicrographs: Pl, plagioclase; Bt, biotite; Kf, K-feldspar; Q, quartz; Px, pyroxene.

(55 vol.%), pyroxene (45 vol.%) and magnetite (5 vol.%) with accessory apatite and baddeleyite. Baddeleyites are brown euhedral small platy prisms with a size ranging from 20 μm to 50 μm.

SD057-4 is a diabase sample collected from the fourth layer sill (Fig. 4B) within the Xiamaling Formation in the Shuangdongzi section in Pingquan (GPS position: E118°45.546′; N40°58.613′). It is composed of plagioclase (55 vol.%), pyroxene (40–45 vol.%) and magnetite (<5 vol.%), with accessory apatite and baddeleyite. Baddeleyites are brown euhedral platy prisms with a size ranging from 50 μm to 100 μm and length/width ratio of 1:1–5:1 (Fig. 7D).

SD069-3 is another diabase sample collected from the third layer sill (Fig. 4A) within the Xiamaling Formation in the Wulongji section in Xiabancheng (GPS position: E118°08.033′; N40°41.159′). It consists of plagioclase (55 vol.%), pyroxene (35–40 vol.%) and magnetite (5–10 vol.%), with accessory apatite and baddeleyite. Baddeleyites are brown euhedral small platy prisms with a size ranging from 30 μm to 60 μm (Fig. 7E).

07014-2 is a biotite alkali feldspar granite collected from the Shangdu-Huade pluton (Fig. 6, GPS position: E113°46.913′; N41°47.042′). It was strongly deformed with a NE-trending foliation and displaying augen structure, quartz recrystallization and

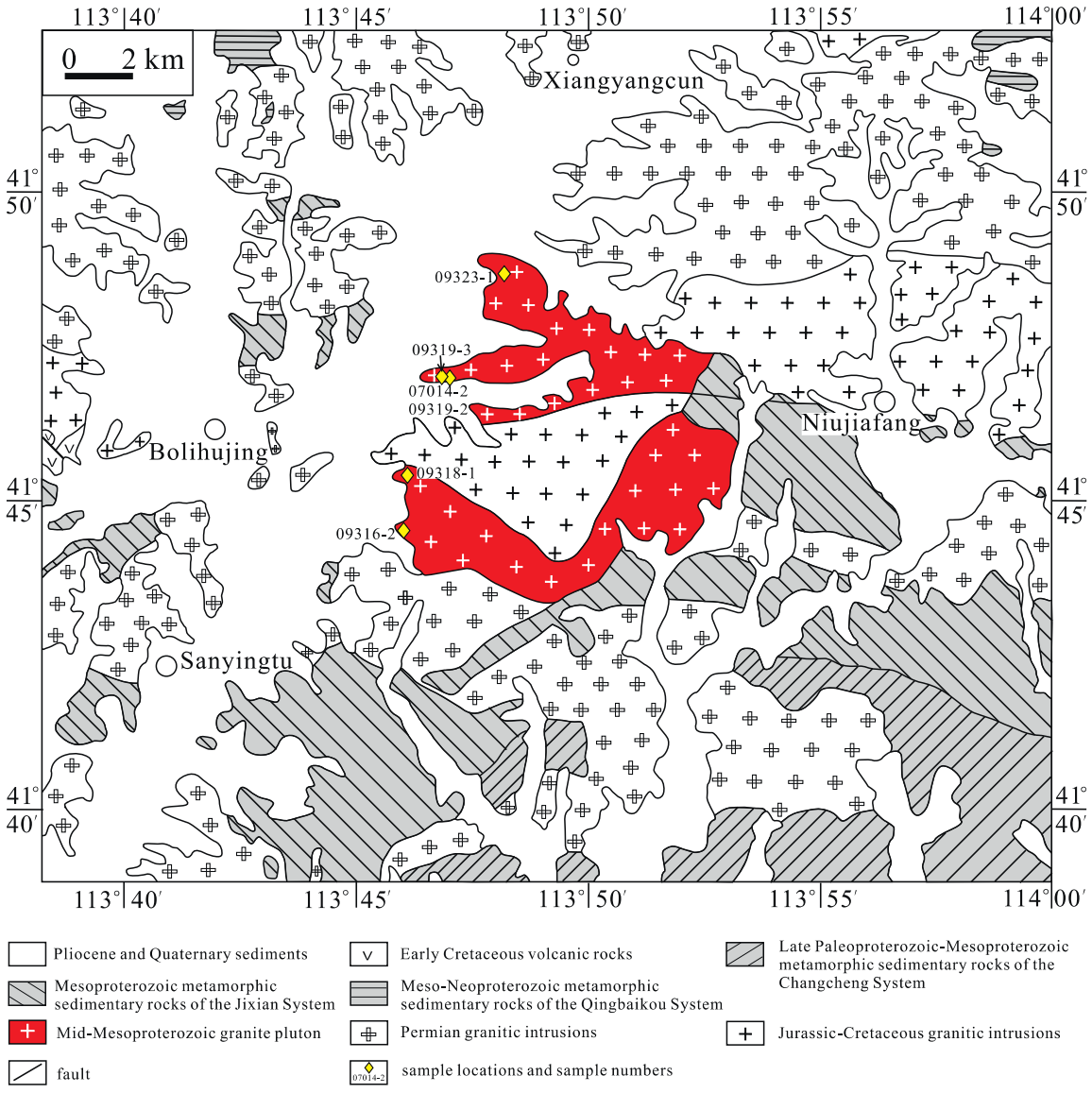


Fig. 6. Geological map of Mid-Mesoproterozoic Shangdu-Huade granite pluton in the northern margin of the NCC (modified after IMBGMR, 1972). Location of this figure is given in Fig. 2 by an open square.

undulose extinction. It consists of K-feldspar (50 vol.%), plagioclase (5 vol.%), quartz (30 vol.%), biotite (10 vol.%) and accessory magnetite, titanite, apatite and zircon and secondary sericite, muscovite (3 vol.%) and epidote (1 vol.%). Zircons from this sample are yellow euhedral prisms with a grain size from 70 μm to 200 μm (Fig. 7F).

Sample 09316-2 is a less deformed biotite monzogranite collected from the Shangdu-Huade pluton (Fig. 6, GPS position: E113°45.898′; N41°44.619′). It displays coarse grained porphyritic texture (Fig. 5J). Different to other granite samples, quartz exhibits no significant undulose extinction or recrystallization. Its main minerals are K-feldspar (25 vol.%), plagioclase (25–30 vol.%), quartz (35 vol.%), biotite (10 vol.%) and accessory magnetite, titanite, apatite and zircon and secondary sericite and muscovite (1–2 vol.%). Zircons are rust yellow subhedral prisms with a grain size from 70 μm to 250 μm (Fig. 7G).

09319-2 is a syenogranite collected from a similar outcrop to sample 07014-2 in the Shangdu-Huade pluton (Fig. 6, GPS position: E113°46.903′; N41°47.044′). It is also strongly deformed with augen structure and NE-trending foliation. It is composed of K-feldspar (60 vol.%), plagioclase (5–10 vol.%), quartz (30 vol.%), biotite (3–5 vol.%) and accessory magnetite, titanite, apatite and zircon. Zircons are rust yellow subhedral prisms with a grain size from 50 μm to 120 μm (Fig. 7H).

09323-1 is another deformed syenogranite collected from the Shangdu-Huade pluton (Fig. 6, GPS position: E113°47.748′; N41°48.567′). Augen structure and NE-trending foliation are very common in the outcrop (Fig. 5I) and quartz displays significant undulose extinction. It consists of K-feldspar (55–60 vol.%), plagioclase (10 vol.%), quartz (30 vol.%), hornblende (3 vol.%), biotite (1 vol.%) and accessory magnetite, titanite, apatite and zircon and secondary sericite and epidote. Zircons are pastel yellow euhedral prisms with a grain size from 70 μm to 200 μm (Fig. 7I).

4. Analytical techniques

4.1. Sample preparation and imaging

Zircons and baddeleyites were separated using conventional crushing and separation techniques and were then handpicked under a binocular microscope. They were mounted in epoxy resin together with zircon and baddeleyite standards and polished to expose the cores of the grains in readiness for photomicrograph

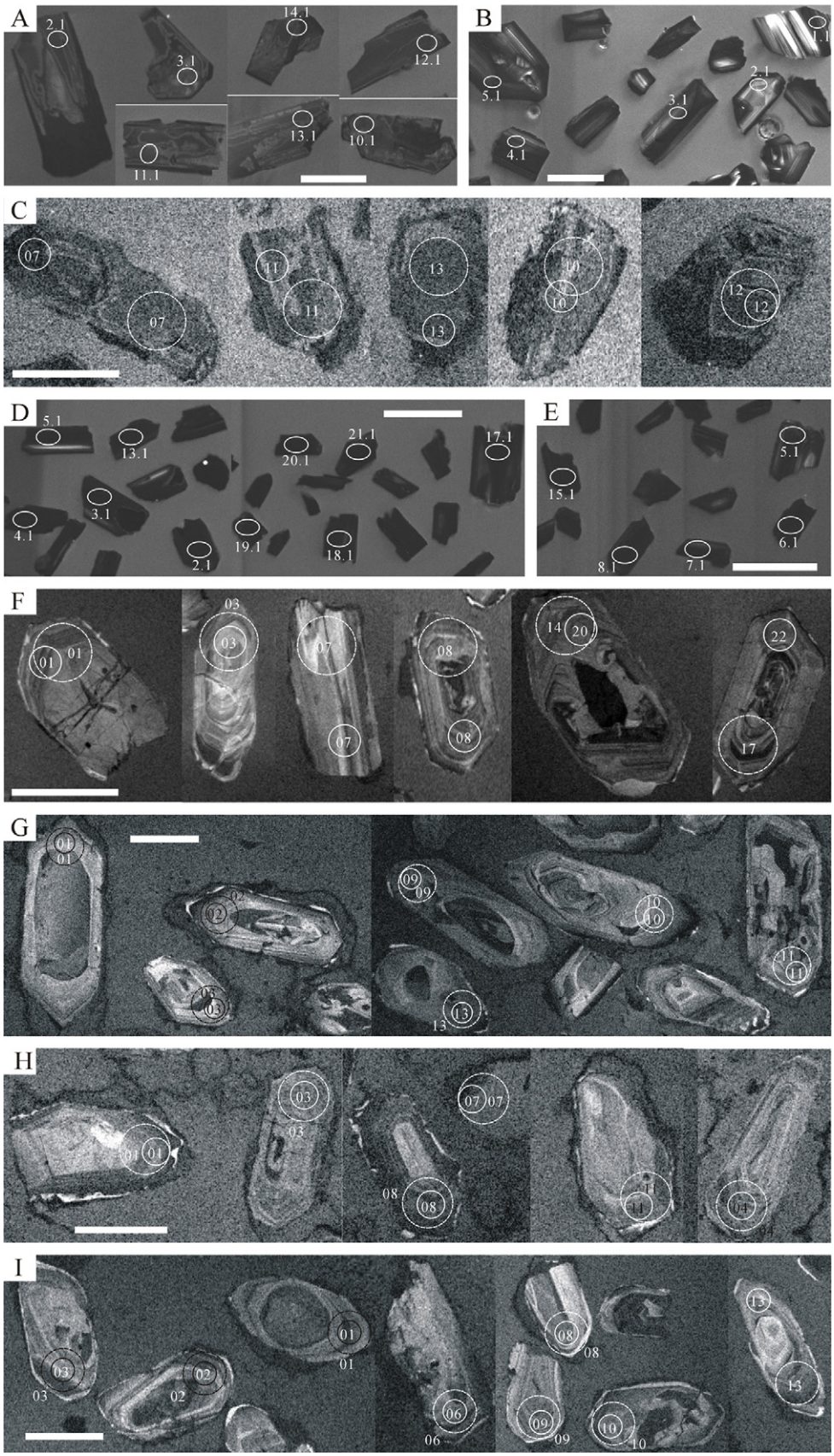


Fig. 7. Representative CL images, with SHRIMP, SIMS and LA-ICP-MS U–Pb and in situ Lu–Hf analyses spots of zircon and baddeleyite grains. (A) 08093-1 (zircon); (B) 08093-1 (baddeleyite); (C) 09053-1 (zircon); (D) SD057-4 (baddeleyite); (E) SD069-3 (baddeleyite); (F) 07014-2 (zircon); (G) 09316-2 (zircon); (H) 09319-2 (zircon); (I) 09323-1 (zircon). Ellipses in (A) and (B), (D)–(E) are SHRIMP and SIMS analysis spots, respectively. Small circles in (C), (G)–(I) are LA-ICP-MS U–Pb analysis spots. Large dashed circles are in situ Lu–Hf analysis spots. Scale bar in each diagram is 100 μm long.

taking, cathodoluminescence (CL) imaging, SIMS, SHRIMP or LA-ICP-MS U–Pb and in situ Lu–Hf isotopic analyses. Zircons and baddeleyites were imaged using the HITACHI S3000-N scanning electron microscope attached with GATAN Chroma CL detector at the Beijing SHRIMP Center and the LEO 1450VP scanning electron microscope attached with GATAN Mini-CL detector at the Institute of Geology and Geophysics (IGG), Chinese Academy of Sciences (CAS), Beijing. CL images of representative zircon and baddeleyite grains are shown in Fig. 7.

4.2. SIMS and LA-ICP-MS U–Pb analysis

Baddeleyite SIMS U–Pb dating was performed on a Cameca IMS-1280 large-radius ion microprobe at the IGG, CAS, Beijing following the method of Li et al. (2009b, 2010c). Oxygen flooding technique is used to improve precision of the analyses (Li et al., 2010d). The ellipsoidal spot is about 20 × 30 μm in size. Each measurement consists of seven cycles, and the total analytical time is ca. 12 min. U–Th–Pb abundances were determined relative to the Harvard standard zircon 91500 (U = 81 ppm, Wiedenbeck et al., 1995). Phalaborwa baddeleyite standard (Heaman and LeCheminant, 1993; Heaman, 2009) is used between every three sample analyses. The measured Pb isotopic compositions were corrected for common Pb using non-radiogenic ^{204}Pb. Concordia diagrams and weighted mean ages were produced using the program ISOPLOT/Ex 3.23 (Ludwig, 2003).

Zircon SHRIMP U–Pb dating was performed on the SHRIMP II at the Beijing SHRIMP Center following the method of Williams (1998). The mass resolution was ca. 5000 at 1% peak height. The spot size of the ion beam was 25–30 μm, and five scans through the mass range were used for data collection. The M257 (561 Ma, U = 840 ppm, Nasdala et al., 2008) and Plešovice (337.1 Ma) zircon standards (Sláma et al., 2008) were used in the analyses for U concentration and age calibration, respectively. Sites for dating were selected on the basis of CL and photomicrograph images. Ages and concordia diagrams were produced using the programs SQUID 1.03 (Ludwig, 2001) and ISOPLOT/Ex 3.23 (Ludwig, 2003).

Zircon LA-ICP-MS U–Pb dating was performed on an excimer (193 nm wavelength) LA-ICP-MS at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences, Wuhan and the State Key Laboratory of Continental Dynamics, Northwest University, Xi'an following the method described by Yuan et al. (2004) and Liu et al. (2010a,b). The ICP-MS used is an Agilent 7500a. The GeoLas 2005 system was used for the laser-ablation experiments. Nitrogen was added into the central gas flow (Ar + He) of the Ar plasma to decrease the detection limit and improve precision (Hu et al., 2008). Sites for dating were selected on the basis of CL and photomicrograph images. The spots used were 30–32 μm in diameter. Common Pb corrections were made using the method described by Andersen (2002). U, Th and Pb concentrations were calibrated by using ^{29}Si as an internal standard and NIST SRM 610 as the reference standard. Isotopic ratios were calculated using GLITTER 4.0 (Macquarie University) and ICPMSDataCal 5.0 (Liu et al., 2010a), which were then corrected for both instrumental mass bias and depth-dependent elemental and isotopic fractionation using Harvard zircon 91500 as an external standard. Concordia diagrams and weighted mean ages were produced using the program ISOPLOT/Ex 3.23 (Ludwig, 2003).

4.3. Major and trace element geochemistry

Major elements were analyzed on fused glass discs by X-ray fluorescence spectrometry at the IGG, CAS, Beijing. Trace elements (including REE) were determined by a Finnigan MAT Element 2 high-resolution ICP-MS at the State Key Laboratory of Mineral Deposits Research, Nanjing University, following the sample preparation and analytical procedures described by Gao et al. (2003). Indium was used as an internal standard to correct for matrix effects and instrument drift. The Chinese national standards GSR-1 (granite) and GSR-3 (basalt) were used to monitor analyses. Errors for major element analysis are within 1%, except for P_2O_5 (5%), and for most trace elements (including REE) are within 10%.

4.4. In-situ Lu–Hf isotope analyses

In-situ Lu–Hf isotope analyses were performed using a Newwave UP213 laser-ablation microprobe attached to a Finnigan Neptune multi-collector ICP-MS at the Institute of Mineral Resources, Chinese Academy of Geological Sciences, Beijing, using techniques and analytical procedures described by Wu et al. (2006) and Hou et al. (2007). The spot diameter is 55 μm. The measured ^{176}Hf/^{177}Hf ratio on the standard zircon (GJ-1) was 0.282009 ± 0.000004 (N = 96), similar to the commonly accepted ^{176}Hf/^{177}Hf ratio of 0.282000 ± 0.000005 measured using the solution method (Morel et al., 2008). For the calculation of initial Hf isotope ratio, the decay constant for ^{176}Lu proposed by Söderlund et al. (2004) was used. For the calculation of $\varepsilon_{Hf}(t)$ values we have adopted the chondritic values of Blichert-Toft and Albarede (1998). Hf model ages (T_{DM} and T_{DM}^{C}) were calculated on the basis of the depleted mantle model described by Griffin et al. (2000, 2004).

4.5. Sm–Nd isotopic analyses

Samples for Sm–Nd isotopic analyses were dissolved in Teflon bombs after being spiked with ^{150}Nd and ^{149}Sm tracers prior to HF + $HClO_4$ dissolution. Sm and Nd were separated using conventional ion exchange procedures and measured using a Finnigan MAT 262 multi-collector mass spectrometer at the IGG, CAS, Beijing. Procedural blanks were <100 pg for Sm and Nd. ^{143}Nd/^{144}Nd was corrected for mass fractionation by normalization to ^{146}Nd/^{144}Nd = 0.7219. Typical within-run precision (2σ) for Nd was estimated to be ±0.000015. The measured values for the JNDi-1 Nd standard were ^{143}Nd/^{144}Nd = 0.512113 ± 5 (N = 5) during the period of data acquisition. The USGS standard BCR-2, prepared using the same procedures as the samples, yielded Sm = 6.73, Nd = 29.40, ^{147}Sm/^{144}Nd = 0.1386 and ^{143}Nd/^{144}Nd = 0.512625 ± 13 (2σ).

5. Results

5.1. U–Pb geochronology

5.1.1. Diabase sill swarms

Although we attempted to separate zircons and baddeleyites from over 30 large samples of diabase (>10 kg) emplaced into different layers (Chuanlinggou, Gaoyuzhuang, Wumishan, Hongshuizhuang, Tieling and Xiamaling Formations) within the Mesoproterozoic sedimentary rocks in different places in the northern NCC, only eight samples yielded sufficient zircon and/or baddeleyite grains for dating. The baddeleyites and/or zircons from five selected diabase samples including two from the sills emplaced into the Wumishan Formation in Chaoyang (samples 08093-1 and 09053-1), one from a sill emplaced into the Tieling Formation in Chaoyang (sample 08100-1), one from a sill emplaced into the Xiamaling Formation in Pingyuan (SD057-4) and one from a sill emplaced into the Xiamaling Formation in Xiabancheng (SD069-3) were analyzed by SHRIMP, SIMS or LA-ICP-MS U–Pb methods and the results were listed in Tables 1 and 2.

Seventeen spots on 17 zircon grains from the Chaoyang diabase sample (08093-1) were measured by the SHRIMP method and the results are plotted on a concordia diagram in Fig. 8A. Most of the analyzed zircons have suffered significant Pb-loss, probably due to

Table 1
SHRIMP and SIMS U–Pb dating results of zircons and baddeleyites from the Mid-Mesoproterozoic magmatic rocks.

Grain spot	$^{206}Pb_c$ (%)	U (ppm)	Th (ppm)	Th/U	$^{204}Pb/^{206}Pb$	$^{207}Pb^*/^{206}Pb^*$	±1σ (%)	$^{207}Pb^*/^{235}U$	±1σ (%)	$^{206}Pb^*/^{238}U$	±1σ (%)	$^{207}Pb^*/^{206}Pb^*$ age (Ma)	±1σ	$^{207}Pb^*/^{235}U$ age (Ma)	±1σ	$^{206}Pb^*/^{238}U$ age (Ma)	±1σ
Zircons from sample 08093-1																	
1.1	0.06	1914	5341	2.88	0.00003	0.08529	0.46	2.489	0.52	0.2116	0.23	1322	9	1269	4	1237	3
2.1	0.22	1195	1618	1.40	0.00013	0.08418	0.95	2.017	0.99	0.1738	0.28	1297	19	1121	7	1033	3
3.1	0.03	2540	7827	3.18	0.00002	0.08507	0.37	2.613	0.43	0.2227	0.23	1317	7	1304	3	1296	3
4.1	1.64	1670	4887	3.02	0.00095	0.07015	4.58	0.475	4.60	0.0491	0.44	933	94	395	15	309	1
5.1	0.07	2009	3700	1.90	0.00004	0.08513	0.46	2.501	0.57	0.2130	0.33	1318	9	1272	4	1245	4
6.1	0.05	2496	5983	2.48	0.00003	0.08468	0.38	2.489	0.45	0.2132	0.24	1308	7	1269	3	1246	3
7.1	0.50	1376	3205	2.41	0.00030	0.08349	1.63	1.017	1.67	0.0883	0.37	1281	32	712	9	545	2
8.1	0.22	2123	6565	3.20	0.00013	0.08551	0.66	2.437	0.74	0.2067	0.33	1327	13	1254	5	1211	4
9.1	1.02	1097	2204	2.08	0.00063	0.08839	1.93	1.568	1.96	0.1287	0.33	1391	37	958	12	780	2
10.1	0.03	2651	6221	2.42	0.00002	0.08463	0.39	2.574	0.47	0.2206	0.27	1307	7	1293	3	1285	3
11.1	0.52	832	3133	3.89	0.00031	0.07956	1.90	0.873	1.95	0.0796	0.43	1186	38	637	9	494	2
12.1	0.10	812	1561	1.99	0.00006	0.08469	0.71	2.565	0.77	0.2196	0.31	1308	14	1291	6	1280	4
13.1	0.09	1083	2065	1.97	0.00005	0.08441	0.64	2.385	0.69	0.2049	0.28	1302	12	1238	5	1202	3
14.1	0.15	1158	2978	2.66	0.00009	0.08358	0.71	2.175	0.77	0.1887	0.28	1283	14	1173	5	1114	3
15.1	0.21	3039	11273	3.83	0.00013	0.07225	2.16	0.640	2.17	0.0643	0.26	993	44	502	9	402	1
16.1	0.08	1486	3332	2.32	0.00005	0.08471	0.74	2.112	0.79	0.1808	0.25	1309	14	1153	5	1071	2
17.1	0.92	2518	7047	2.89	0.00055	0.08066	5.17	0.880	5.18	0.0791	0.32	1213	102	641	25	491	2
Baddeleyites from sample 08093-1																	
1.1	0.00	257	11	0.04	0.00000	0.08547	0.64	2.754	2.94	0.2337	2.87	1326	12	1343	22	1354	35
2.1	0.00	170	3	0.02	0.00000	0.08581	1.17	2.727	3.11	0.2305	2.88	1334	22	1336	23	1337	35
3.1	0.00	858	67	0.08	0.00000	0.08576	0.39	2.945	2.90	0.2491	2.88	1333	8	1394	22	1434	37
4.1	0.01	590	20	0.03	0.00001	0.08571	0.40	2.752	2.90	0.2328	2.87	1332	8	1343	22	1349	35
5.1	0.00	275	5	0.02	0.00000	0.08624	0.74	2.830	2.96	0.2380	2.87	1344	14	1364	22	1376	36
6.1	0.01	243	4	0.02	0.00001	0.08496	0.66	2.819	2.96	0.2407	2.88	1315	13	1361	22	1390	36
7.1	0.00	238	11	0.05	0.00000	0.08389	1.02	2.603	3.05	0.2250	2.87	1290	20	1301	23	1308	34
8.1	0.00	226	7	0.03	0.00000	0.08489	0.67	2.784	2.95	0.2378	2.87	1313	13	1351	22	1375	36
9.1	0.01	352	16	0.05	0.00000	0.08512	0.55	2.793	2.93	0.2379	2.88	1318	11	1354	22	1376	36
10.1	0.03	174	4	0.02	0.00002	0.08590	0.99	2.631	3.04	0.2222	2.87	1336	19	1309	23	1293	34
11.1	0.00	181	8	0.04	0.00000	0.08644	1.08	2.670	3.07	0.2240	2.87	1348	21	1320	23	1303	34
12.1	0.00	127	2	0.02	0.00000	0.08515	0.93	2.767	3.03	0.2356	2.88	1319	18	1347	23	1364	36
13.1	0.00	208	10	0.05	0.00000	0.08446	0.77	2.792	3.00	0.2397	2.90	1303	15	1353	23	1385	36
14.1	0.02	755	65	0.09	0.00001	0.08576	0.38	2.870	2.90	0.2427	2.87	1333	7	1374	22	1401	36
15.1	0.00	447	21	0.05	0.00000	0.08508	0.48	2.653	2.92	0.2261	2.88	1317	9	1315	22	1314	34
16.1	0.00	297	21	0.07	0.00000	0.08590	0.65	2.769	2.95	0.2338	2.88	1336	12	1347	22	1354	35
17.1	0.00	179	6	0.03	0.00000	0.08527	0.74	2.790	2.97	0.2373	2.88	1322	14	1353	22	1373	36
18.1	0.03	96	3	0.03	0.00001	0.08368	1.15	2.785	3.10	0.2414	2.88	1285	22	1351	23	1394	36
19.1	0.01	866	42	0.05	0.00000	0.08490	0.34	2.689	2.89	0.2297	2.87	1313	7	1326	22	1333	35
20.1	0.00	548	49	0.09	0.00000	0.08553	0.64	2.830	2.94	0.2400	2.87	1328	12	1363	22	1386	36
Baddeleyites from sample 08100-1																	
1.1	0.20	828	50	0.06	0.00011	0.08553	0.49	2.851	2.91	0.2417	2.87	1328	9	1369	22	1396	36
2.1	0.14	284	10	0.04	0.00007	0.08449	0.84	2.565	2.99	0.2202	2.87	1304	16	1291	22	1283	33
3.1	0.38	458	32	0.07	0.00020	0.08444	0.66	2.741	2.95	0.2354	2.87	1303	13	1340	22	1363	35

Table 1 (*Continued*)

Grain spot	$^{206}Pb_c$ (%)	U (ppm)	Th (ppm)	Th/U	$^{204}Pb/^{206}Pb$	$^{207}Pb^*/^{206}Pb^*$	±1σ (%)	$^{207}Pb^*/^{235}U$	±1σ (%)	$^{206}Pb^*/^{238}U$	±1σ (%)	$^{207}Pb^*/^{206}Pb^*$ age (Ma)	±1σ	$^{207}Pb^*/^{235}U$ age (Ma)	±1σ	$^{206}Pb^*/^{238}U$ age (Ma)	±1σ
Baddeleyites from sample SD057-4																	
1.1	0.00	1039	140	0.13	0.00000	0.08529	0.34	3.022	2.90	0.2570	2.88	1322	7	1413	22	1474	38
2.1	0.00	2493	552	0.22	0.00000	0.08563	0.38	3.112	2.98	0.2636	2.96	1330	7	1436	23	1508	40
3.1	1.13	804	50	0.06	0.00061	0.08524	6.08	2.977	6.73	0.2533	2.87	1321	114	1402	52	1456	38
4.1	0.02	706	36	0.05	0.00001	0.08549	0.38	2.845	2.90	0.2414	2.88	1327	7	1368	22	1394	36
5.1	0.00	594	38	0.06	0.00000	0.08574	0.41	2.825	2.90	0.2390	2.87	1332	8	1362	22	1381	36
6.1	0.01	709	82	0.12	0.00001	0.08497	0.41	2.977	2.90	0.2541	2.87	1315	8	1402	22	1460	38
7.1	0.02	1168	192	0.16	0.00001	0.08501	0.30	3.198	2.89	0.2728	2.87	1316	6	1457	23	1555	40
8.1	0.00	1562	193	0.12	0.00000	0.08516	0.25	3.075	2.95	0.2619	2.94	1319	5	1426	23	1499	39
9.1	0.02	470	34	0.07	0.00001	0.08576	0.61	2.825	2.93	0.2389	2.87	1333	12	1362	22	1381	36
10.1	0.15	466	36	0.08	0.00008	0.08476	0.51	2.878	2.95	0.2462	2.91	1310	10	1376	23	1419	37
11.1	0.46	212	18	0.08	0.00025	0.08570	1.14	2.885	3.10	0.2442	2.88	1331	22	1378	24	1408	37
12.1	0.01	703	72	0.10	0.00001	0.08548	0.41	2.953	2.91	0.2506	2.88	1326	8	1396	22	1441	37
13.1	0.00	434	38	0.09	0.00000	0.08482	0.48	2.943	2.93	0.2516	2.89	1311	9	1393	22	1447	38
14.1	0.00	1172	294	0.25	0.00000	0.08574	0.44	3.139	2.91	0.2655	2.87	1332	8	1442	23	1518	39
15.1	0.54	647	35	0.05	0.00029	0.08478	0.67	2.784	2.95	0.2382	2.87	1310	13	1351	22	1377	36
16.1	0.03	323	11	0.04	0.00002	0.08531	0.56	2.990	3.05	0.2542	3.00	1323	11	1405	23	1460	39
17.1	0.01	658	72	0.11	0.00000	0.08541	0.37	2.899	2.90	0.2461	2.87	1325	7	1382	22	1418	37
18.1	0.00	774	76	0.10	0.00000	0.08530	0.65	2.599	2.95	0.2209	2.87	1322	13	1300	22	1287	34
19.1	0.02	555	22	0.04	0.00001	0.08452	0.44	2.898	2.91	0.2487	2.88	1304	9	1381	22	1432	37
20.1	0.07	645	40	0.06	0.00004	0.08485	0.46	2.905	2.91	0.2483	2.87	1312	9	1383	22	1430	37
21.1	0.26	324	9	0.03	0.00014	0.08358	0.82	2.904	2.99	0.2520	2.87	1283	16	1383	23	1449	37
Baddeleyites from sample SD069-3																	
1.1	0.00	1332	142	0.11	0.00000	0.08531	0.27	2.995	2.90	0.2546	2.88	1323	5	1406	22	1462	38
2.1	0.34	376	25	0.07	0.00018	0.08607	0.87	2.755	3.03	0.2322	2.91	1340	17	1344	23	1346	35
3.1	0.01	1997	221	0.11	0.00001	0.08558	0.33	3.017	2.91	0.2557	2.89	1329	6	1412	22	1468	38
4.1	0.11	202	27	0.13	0.00006	0.08591	1.03	2.909	3.05	0.2456	2.87	1336	20	1384	23	1416	37
5.1	0.00	261	30	0.11	0.00000	0.08617	0.65	2.901	2.94	0.2442	2.87	1342	12	1382	22	1408	36
6.1	0.25	651	73	0.11	0.00013	0.08623	0.49	2.998	2.98	0.2522	2.94	1343	9	1407	23	1450	38
7.1	0.09	311	13	0.04	0.00005	0.08482	0.64	2.835	2.94	0.2424	2.87	1312	12	1365	22	1399	36
8.1	0.00	1165	170	0.15	0.00000	0.08503	0.49	3.028	2.91	0.2583	2.87	1316	9	1415	22	1481	38
9.1	0.00	1130	74	0.07	0.00000	0.08592	0.57	3.197	2.93	0.2699	2.87	1336	11	1456	23	1540	39
10.1	0.01	1842	64	0.03	0.00000	0.08526	0.28	2.832	2.89	0.2409	2.87	1321	5	1364	22	1392	36
11.1	0.24	1220	233	0.19	0.00013	0.08524	0.62	2.894	2.94	0.2463	2.87	1321	12	1380	22	1419	37
12.1	0.01	1217	132	0.11	0.00001	0.08539	0.30	2.937	2.89	0.2494	2.87	1324	6	1391	22	1435	37
13.1	0.01	740	113	0.15	0.00001	0.08490	0.39	2.940	2.91	0.2511	2.88	1313	7	1392	22	1444	37
14.1	0.05	749	110	0.15	0.00003	0.08612	0.45	2.963	2.92	0.2496	2.88	1341	9	1398	22	1436	37
15.1	0.02	344	22	0.06	0.00001	0.08493	0.56	2.670	2.93	0.2280	2.87	1314	11	1320	22	1324	34

Errors are 1σ; Pb_c and Pb^* indicate the common and radiogenic portions, respectively; common Pb corrected using measured ^{204}Pb.

Table 2
LA-ICP-MS U–Pb dating results of zircons from the Mid-Mesoproterozoic magmatic rocks.

Grain-spot	Pb (ppm)	^{232}Th (ppm)	^{238}U (ppm)	^{232}Th/^{238}U	^{204}Pb/^{206}Pb	^{207}Pb*/^{206}Pb*	^{207}Pb*/^{235}U	^{206}Pb*/^{238}U	^{207}Pb*/^{206}Pb* age (Ma)	^{207}Pb*/^{235}U age (Ma)	^{206}Pb*/^{238}U age (Ma)
Sample 09053-1 (diabase)											
01	123	414	351	1.18	<0.00033	0.0847 ± 0.0010	2.816 ± 0.036	0.2406 ± 0.0015	1309 ± 24	1360 ± 9	1390 ± 8
02	175	716	494	1.45	0.00017	0.0861 ± 0.0013	2.684 ± 0.045	0.2253 ± 0.0017	1340 ± 29	1324 ± 12	1310 ± 9
03	368	1908	1097	1.74	0.00002	0.0846 ± 0.0011	2.389 ± 0.037	0.2040 ± 0.0019	1307 ± 24	1239 ± 11	1197 ± 10
04	157	616	439	1.40	0.00007	0.0856 ± 0.0011	2.772 ± 0.037	0.2341 ± 0.0014	1328 ± 25	1348 ± 10	1356 ± 7
05	120	422	353	1.20	0.00018	0.0862 ± 0.0012	2.775 ± 0.040	0.2328 ± 0.0014	1343 ± 27	1349 ± 11	1349 ± 8
06	200	1250	865	1.45	0.00018	0.0843 ± 0.0012	1.734 ± 0.027	0.1485 ± 0.0013	1302 ± 27	1021 ± 10	892 ± 7
07	236	1132	742	1.53	0.00016	0.0857 ± 0.0012	2.550 ± 0.067	0.2136 ± 0.0041	1332 ± 27	1287 ± 19	1248 ± 22
08	339	1443	872	1.65	0.00014	0.0851 ± 0.0009	2.802 ± 0.032	0.2378 ± 0.0013	1320 ± 21	1356 ± 9	1375 ± 7
09	178	187	687	0.27	<0.00016	0.0849 ± 0.0010	2.629 ± 0.036	0.2238 ± 0.0018	1322 ± 18	1309 ± 10	1302 ± 9
10	190	820	607	1.35	0.00034	0.0857 ± 0.0013	2.394 ± 0.034	0.2020 ± 0.0015	1331 ± 29	1241 ± 10	1186 ± 8
11	385	1833	999	1.83	0.00001	0.0848 ± 0.0011	2.635 ± 0.039	0.2250 ± 0.0022	1310 ± 26	1311 ± 11	1308 ± 12
12	277	1161	765	1.52	0.00000	0.0858 ± 0.0012	2.698 ± 0.037	0.2273 ± 0.0017	1333 ± 27	1328 ± 10	1320 ± 9
13	124	483	340	1.42	0.00024	0.0845 ± 0.0022	2.753 ± 0.056	0.2368 ± 0.0019	1306 ± 50	1343 ± 15	1370 ± 10
14	231	953	641	1.49	0.00010	0.0839 ± 0.0009	2.620 ± 0.039	0.2254 ± 0.0024	1300 ± 22	1306 ± 11	1310 ± 13
15	267	857	929	0.92	0.00004	0.0860 ± 0.0013	2.429 ± 0.038	0.2043 ± 0.0023	1339 ± 28	1251 ± 11	1198 ± 13
16	187	518	598	0.87	0.00006	0.0859 ± 0.0013	2.655 ± 0.040	0.2234 ± 0.0020	1337 ± 28	1316 ± 11	1300 ± 11
17	211	1560	539	2.90	0.00018	0.0860 ± 0.0010	2.269 ± 0.043	0.1899 ± 0.0027	1339 ± 22	1203 ± 13	1121 ± 14
18	351	1673	999	1.68	0.00011	0.0856 ± 0.0009	2.633 ± 0.034	0.2222 ± 0.0019	1331 ± 20	1310 ± 9	1294 ± 10
19	252	931	685	1.36	<0.00015	0.0847 ± 0.0010	2.794 ± 0.032	0.2387 ± 0.0013	1309 ± 21	1354 ± 9	1380 ± 7
20	91.0	402	294	1.37	0.00029	0.0871 ± 0.0023	2.399 ± 0.062	0.1994 ± 0.0025	1362 ± 51	1242 ± 19	1172 ± 13
Sample 07014-2 (granite)											
01	83.4	36	351	0.10	0.00055	0.0858 ± 0.0018	2.727 ± 0.027	0.2306 ± 0.0021	1334 ± 39	1336 ± 7	1338 ± 11
02	53.3	57	195	0.29	0.00159	0.0938 ± 0.0019	3.239 ± 0.033	0.2507 ± 0.0023	1503 ± 39	1467 ± 8	1442 ± 12
03	51.6	110	186	0.59	0.00070	0.0868 ± 0.0018	2.862 ± 0.030	0.2393 ± 0.0023	1356 ± 40	1372 ± 8	1383 ± 12
04	52.6	81	207	0.39	0.00124	0.0891 ± 0.0019	2.897 ± 0.030	0.2359 ± 0.0022	1406 ± 39	1381 ± 8	1365 ± 11
05	116	53	529	0.10	0.00059	0.0848 ± 0.0018	2.476 ± 0.025	0.2120 ± 0.0020	1310 ± 40	1265 ± 7	1239 ± 10
06	170	60	558	0.11	0.00046	0.1048 ± 0.0021	4.184 ± 0.040	0.2896 ± 0.0027	1711 ± 37	1671 ± 8	1639 ± 13
07	65.7	120	240	0.50	0.00170	0.0858 ± 0.0018	2.839 ± 0.029	0.2400 ± 0.0022	1334 ± 40	1366 ± 8	1387 ± 12
08	118	66	529	0.13	0.00058	0.0848 ± 0.0017	2.501 ± 0.024	0.2141 ± 0.0020	1310 ± 39	1272 ± 7	1251 ± 10
09	124	66	404	0.16	0.00118	0.1059 ± 0.0022	4.191 ± 0.042	0.2872 ± 0.0027	1729 ± 37	1672 ± 8	1628 ± 13
10	88.3	75	341	0.22	0.00134	0.0853 ± 0.0018	2.551 ± 0.026	0.2171 ± 0.0020	1321 ± 40	1287 ± 8	1266 ± 11
11	90.3	178	332	0.54	0.00228	0.0978 ± 0.0016	2.638 ± 0.066	0.1913 ± 0.0030	1583 ± 31	1311 ± 18	1129 ± 16
12	143	56	554	0.10	0.00108	0.0857 ± 0.0012	2.747 ± 0.062	0.2288 ± 0.0036	1331 ± 27	1341 ± 17	1328 ± 19
13	105	74	369	0.20	0.00127	0.0872 ± 0.0014	2.840 ± 0.051	0.2337 ± 0.0028	1365 ± 29	1366 ± 14	1354 ± 14
14	146	109	408	0.27	0.00090	0.1078 ± 0.0016	4.361 ± 0.078	0.2901 ± 0.0039	1763 ± 28	1705 ± 15	1642 ± 19
15	75.3	109	201	0.54	0.00162	0.0971 ± 0.0017	3.895 ± 0.085	0.2873 ± 0.0045	1569 ± 32	1613 ± 18	1628 ± 23
16	191	265	508	0.52	0.00055	0.1044 ± 0.0014	4.231 ± 0.077	0.2898 ± 0.0035	1706 ± 25	1680 ± 15	1641 ± 18
17	111	39	449	0.09	0.00098	0.0851 ± 0.0015	2.578 ± 0.054	0.2187 ± 0.0031	1318 ± 33	1294 ± 15	1275 ± 17
18	92.8	113	332	0.34	0.00103	0.0876 ± 0.0010	2.765 ± 0.057	0.2255 ± 0.0022	1376 ± 22	1346 ± 15	1311 ± 11
19	115	42	461	0.09	0.00081	0.0852 ± 0.0009	2.506 ± 0.055	0.2107 ± 0.0016	1320 ± 19	1274 ± 16	1233 ± 8
20	80.2	29	339	0.09	0.00101	0.0867 ± 0.0039	2.448 ± 0.106	0.2034 ± 0.0020	1354 ± 87	1257 ± 31	1193 ± 10
21	109	237	399	0.59	0.00077	0.0829 ± 0.0007	2.393 ± 0.052	0.2067 ± 0.0012	1278 ± 17	1240 ± 15	1211 ± 6
22	136	219	590	0.37	0.00066	0.0862 ± 0.0008	2.314 ± 0.048	0.1931 ± 0.0022	1343 ± 18	1217 ± 15	1138 ± 12
23	126	43	483	0.09	0.00059	0.0849 ± 0.0007	2.649 ± 0.052	0.2239 ± 0.0012	1322 ± 17	1314 ± 15	1302 ± 6
24	156	139	496	0.28	0.00055	0.0983 ± 0.0006	3.404 ± 0.069	0.2488 ± 0.0018	1594 ± 13	1505 ± 16	1432 ± 9
25	114	131	413	0.32	0.00072	0.0860 ± 0.0006	2.685 ± 0.050	0.2250 ± 0.0014	1339 ± 13	1324 ± 14	1308 ± 7
26	80.3	55	295	0.18	0.00086	0.0869 ± 0.0008	2.749 ± 0.048	0.2281 ± 0.0013	1359 ± 17	1342 ± 13	1325 ± 7
27	124	254	449	0.57	0.00061	0.0871 ± 0.0007	2.575 ± 0.043	0.2131 ± 0.0010	1365 ± 21	1294 ± 12	1245 ± 5
28	133	296	471	0.63	0.00048	0.0857 ± 0.0007	2.554 ± 0.044	0.2148 ± 0.0010	1331 ± 15	1288 ± 13	1254 ± 5
29	220	576	709	0.81	0.00045	0.0852 ± 0.0005	2.650 ± 0.047	0.2244 ± 0.0012	1320 ± 10	1315 ± 13	1305 ± 6
30	85.3	61	302	0.20	0.00097	0.0847 ± 0.0009	2.768 ± 0.061	0.2355 ± 0.0014	1309 ± 20	1347 ± 16	1363 ± 7

Table 2 (*Continued*)

Grain-spot	Pb (ppm)	^{232}Th (ppm)	^{238}U (ppm)	^{232}Th/^{238}U	^{204}Pb/^{206}Pb	^{207}Pb*/^{206}Pb*	^{207}Pb*/^{235}U	^{206}Pb*/^{238}U	^{207}Pb*/^{206}Pb* age (Ma)	^{207}Pb*/^{235}U age (Ma)	^{206}Pb*/^{238}U age (Ma)
Sample 09316-2 (granite)											
01	103	83	310	0.27	0.00002	0.0999 ± 0.0012	3.754 ± 0.044	0.2719 ± 0.0015	1621 ± 22	1583 ± 9	1550 ± 8
02	86.9	75	286	0.26	<0.00022	0.0952 ± 0.0013	3.406 ± 0.066	0.2566 ± 0.0027	1532 ± 30	1506 ± 15	1473 ± 14
03	80.8	68	302	0.22	0.00000	0.0848 ± 0.0011	2.723 ± 0.037	0.2321 ± 0.0015	1322 ± 25	1335 ± 10	1346 ± 8
04	75.5	193	157	1.23	<0.00033	0.1105 ± 0.0015	4.932 ± 0.068	0.3229 ± 0.0018	1809 ± 25	1808 ± 12	1804 ± 9
05	84.7	41	323	0.13	0.00016	0.0862 ± 0.0014	2.748 ± 0.043	0.2307 ± 0.0022	1344 ± 27	1342 ± 12	1338 ± 12
06	189	57	604	0.09	0.00005	0.0950 ± 0.0012	3.583 ± 0.047	0.2725 ± 0.0015	1529 ± 24	1546 ± 10	1554 ± 7
07	48.6	92	161	0.57	0.00032	0.0839 ± 0.0013	2.740 ± 0.043	0.2361 ± 0.0013	1300 ± 31	1339 ± 12	1367 ± 7
08	132	78	507	0.15	0.00014	0.0847 ± 0.0012	2.643 ± 0.035	0.2254 ± 0.0013	1310 ± 26	1313 ± 10	1311 ± 7
09	46.8	56	192	0.29	<0.00035	0.0846 ± 0.0016	2.406 ± 0.049	0.2055 ± 0.0020	1306 ± 33	1244 ± 14	1205 ± 10
10	80.2	46	303	0.15	<0.00024	0.0845 ± 0.0010	2.685 ± 0.033	0.2295 ± 0.0013	1306 ± 24	1324 ± 9	1332 ± 7
11	106	82	386	0.21	0.00005	0.0839 ± 0.0010	2.736 ± 0.035	0.2355 ± 0.0018	1300 ± 22	1338 ± 10	1363 ± 9
12	235	138	628	0.22	0.00008	0.1064 ± 0.0011	4.561 ± 0.049	0.3095 ± 0.0017	1739 ± 23	1742 ± 9	1738 ± 8
13	65.6	56	238	0.23	0.00007	0.0839 ± 0.0011	2.718 ± 0.034	0.2342 ± 0.0015	1300 ± 26	1333 ± 9	1357 ± 8
14	241	171	605	0.28	0.00001	0.1080 ± 0.0011	4.876 ± 0.063	0.3253 ± 0.0025	1766 ± 20	1798 ± 11	1816 ± 12
15	78.2	57	274	0.21	<0.00022	0.0856 ± 0.0011	2.893 ± 0.036	0.2439 ± 0.0014	1331 ± 24	1380 ± 9	1407 ± 7
16	104	113	238	0.47	0.00002	0.1154 ± 0.0014	5.474 ± 0.072	0.3422 ± 0.0022	1887 ± 22	1897 ± 11	1897 ± 11
Sample 09319-2 (granite)											
01	130	43	507	0.09	0.00011	0.0855 ± 0.0010	2.722 ± 0.033	0.2296 ± 0.0017	1328 ± 22	1334 ± 9	1333 ± 9
02	47.1	69	105	0.66	<0.00049	0.1105 ± 0.0014	5.687 ± 0.093	0.3721 ± 0.0044	1807 ± 24	1929 ± 14	2039 ± 21
03	117	44	462	0.09	0.00001	0.0848 ± 0.0010	2.656 ± 0.031	0.2260 ± 0.0011	1322 ± 56	1316 ± 8	1314 ± 6
04	120	42	465	0.09	0.00017	0.0845 ± 0.0010	2.703 ± 0.033	0.2308 ± 0.0012	1306 ± 24	1329 ± 9	1339 ± 6
05	90.5	30	361	0.08	0.00013	0.0863 ± 0.0012	2.710 ± 0.039	0.2267 ± 0.0013	1346 ± 28	1331 ± 11	1317 ± 7
06	163	72	471	0.15	0.00006	0.1090 ± 0.0013	4.579 ± 0.057	0.3033 ± 0.0018	1783 ± 21	1745 ± 10	1707 ± 9
07	89.8	44	363	0.12	<0.00027	0.0863 ± 0.0010	2.645 ± 0.031	0.2213 ± 0.0011	1346 ± 22	1313 ± 9	1289 ± 6
08	104	47	399	0.12	0.00013	0.0839 ± 0.0010	2.728 ± 0.035	0.2349 ± 0.0014	1300 ± 18	1336 ± 9	1360 ± 8
09	159	49	590	0.08	0.00004	0.0857 ± 0.0010	2.868 ± 0.033	0.2420 ± 0.0013	1331 ± 22	1374 ± 9	1397 ± 7
10	114	45	455	0.10	0.00008	0.0862 ± 0.0011	2.658 ± 0.034	0.2230 ± 0.0012	1344 ± 25	1317 ± 9	1298 ± 6
11	95.4	100	378	0.26	0.00004	0.0861 ± 0.0011	2.589 ± 0.034	0.2173 ± 0.0012	1343 ± 25	1298 ± 10	1268 ± 6
12	103	154	410	0.37	0.00011	0.0870 ± 0.0018	2.489 ± 0.055	0.2068 ± 0.0013	1361 ± 40	1269 ± 16	1212 ± 7
13	104	47	406	0.12	0.00001	0.0849 ± 0.0010	2.661 ± 0.031	0.2267 ± 0.0012	1315 ± 22	1318 ± 9	1317 ± 6
14	77.1	30	311	0.10	0.00009	0.0847 ± 0.0011	2.625 ± 0.034	0.2245 ± 0.0014	1309 ± 58	1308 ± 10	1306 ± 7
15	97.4	55	396	0.14	0.00009	0.0830 ± 0.0012	2.489 ± 0.040	0.2171 ± 0.0017	1278 ± 30	1269 ± 12	1266 ± 9
16	70.5	76	236	0.32	0.00019	0.0930 ± 0.0014	3.346 ± 0.067	0.2590 ± 0.0031	1488 ± 28	1492 ± 16	1485 ± 16
Sample 09323-1 (granite)											
01	123	34	478	0.07	0.00021	0.0861 ± 0.0011	2.744 ± 0.040	0.2304 ± 0.0019	1340 ± 24	1340 ± 11	1337 ± 10
02	151	74	578	0.13	0.00004	0.0851 ± 0.0009	2.673 ± 0.029	0.2272 ± 0.0012	1317 ± 20	1321 ± 8	1320 ± 6
03	114	171	393	0.43	<0.00024	0.0862 ± 0.0010	2.809 ± 0.035	0.2358 ± 0.0016	1343 ± 22	1358 ± 9	1365 ± 8
04	131	232	471	0.49	<0.00023	0.0870 ± 0.0010	2.690 ± 0.034	0.2234 ± 0.0014	1361 ± 22	1326 ± 9	1300 ± 7
05	242	308	809	0.38	0.00028	0.0899 ± 0.0011	3.054 ± 0.042	0.2453 ± 0.0016	1433 ± 24	1421 ± 10	1414 ± 8
06	89.5	165	330	0.50	<0.00026	0.0854 ± 0.0015	2.560 ± 0.047	0.2166 ± 0.0018	1326 ± 29	1289 ± 13	1264 ± 10
07	303	284	1032	0.28	0.00002	0.0854 ± 0.0011	2.934 ± 0.037	0.2482 ± 0.0014	1325 ± 24	1391 ± 9	1429 ± 7
08	129	79	499	0.16	0.00024	0.0860 ± 0.0011	2.665 ± 0.035	0.2238 ± 0.0014	1339 ± 23	1319 ± 10	1302 ± 8
09	145	84	554	0.15	0.00010	0.0859 ± 0.0010	2.705 ± 0.033	0.2276 ± 0.0012	1337 ± 19	1330 ± 9	1322 ± 6
10	97.4	178	314	0.57	0.00014	0.0856 ± 0.0011	2.874 ± 0.037	0.2426 ± 0.0016	1331 ± 24	1375 ± 10	1400 ± 8
11	105	200	356	0.56	0.00020	0.0851 ± 0.0010	2.722 ± 0.035	0.2313 ± 0.0015	1318 ± 23	1335 ± 10	1342 ± 8
12	460	748	1422	0.53	0.00004	0.0848 ± 0.0009	3.037 ± 0.037	0.2586 ± 0.0019	1322 ± 20	1417 ± 9	1483 ± 10
13	170	168	604	0.28	<0.00014	0.0860 ± 0.0010	2.784 ± 0.033	0.2340 ± 0.0013	1339 ± 21	1351 ± 9	1355 ± 7
14	90.7	191	302	0.63	0.00021	0.0846 ± 0.0011	2.702 ± 0.037	0.2309 ± 0.0013	1306 ± 26	1329 ± 10	1339 ± 7
15	63.9	155	213	0.73	0.00005	0.0852 ± 0.0013	2.642 ± 0.039	0.2241 ± 0.0014	1321 ± 29	1312 ± 11	1304 ± 7
16	144	220	480	0.46	0.00022	0.0846 ± 0.0012	2.824 ± 0.041	0.2412 ± 0.0014	1306 ± 27	1362 ± 11	1393 ± 7

Errors are 1σ; Pb refers to total Pb; Pb* indicate the radiogenic lead portion; common Pb corrected using method described by Andersen (2002).

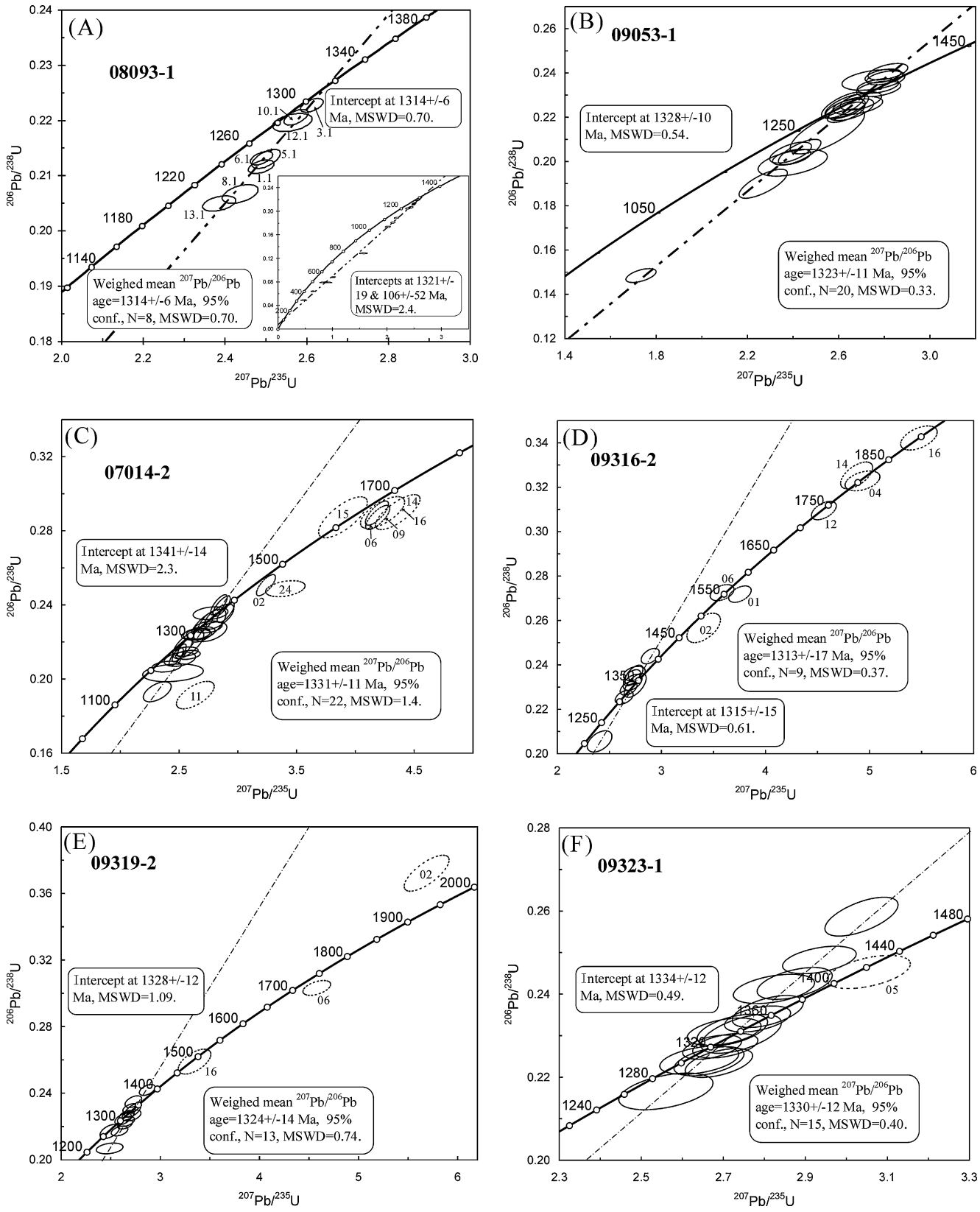


Fig. 8. U–Pb concordia diagrams for zircons from Mid-Mesoproterozoic magmatic rocks. Data-point error crosses are 2σ.

their high contents of Th and U and later thermal disturbance. All the analyzed spots are aligned near a discordia line with upper and lower intercept ages of 1321 ± 19 Ma and 106 ± 52 Ma, respectively (Fig. 8A inset). Except for nine spots (2.1, 4.1, 7.1, 9.1, 11.1, 14.1, 15.1, 16.1, 17.1) that show significant Pb-loss, the remaining spots yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1314 ± 6 Ma (95% confidence, MSWD = 0.70, N = 8, Fig. 8A). This age is exactly the same as that of the upper intercept age of these analyses (1314 ± 6 Ma,

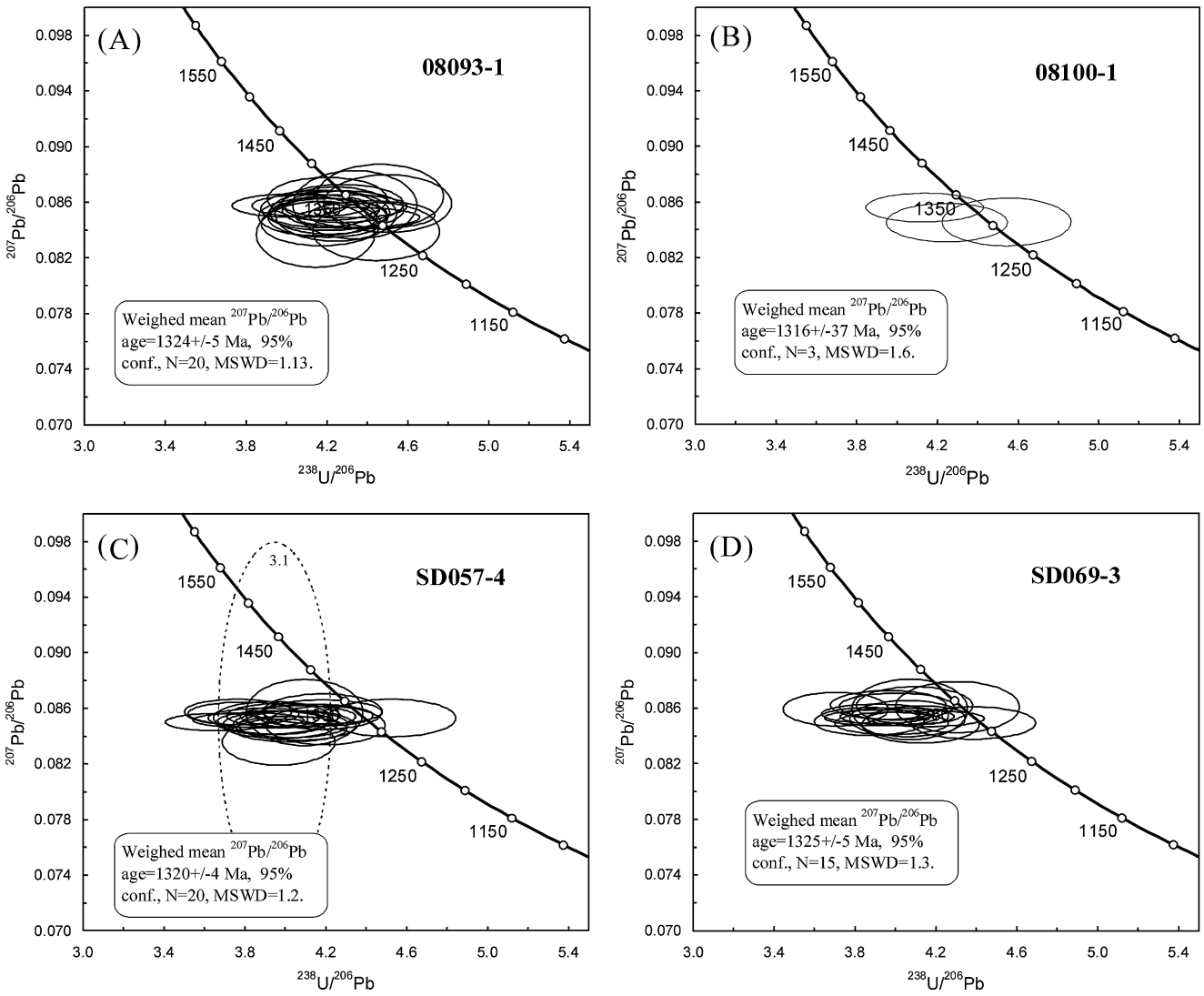


Fig. 9. U–Pb concordia diagrams for baddeleyites from Mid-Mesoproterozoic diabase sills. Data-point error crosses are 2σ.

MSWD = 0.70; the lower intercept was forced through 0 Ma). High Th/U ratios (1.38–3.92) and euhedral prismatic morphology of these zircons (Fig. 7A) suggest they are magmatic in origin. Metamictization in zircons can be observed in their CL images (Fig. 7A). Therefore, the upper intercept age of 1321 ± 19 Ma is interpreted as the emplacement age of the Chaoyang diabase sills.

The SIMS U–Pb analytical results of 20 baddeleyite grains from the Chaoyang diabase sample (08093-1) are listed in Table 1 and plotted on a concordia diagram in Fig. 9A. The analyzed $^{207}Pb/^{206}Pb$ ages are well concentrated and all the analyses yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1324 ± 5 Ma (95% confidence, MSWD = 1.13, N = 20). Because baddeleyites in mafic rocks are all magmatic in origin and contain almost no inheritance (Krogh et al., 1987; Heaman and LeCheminant, 1993; Wingate, 2001), the mean $^{207}Pb/^{206}Pb$ age of 1324 ± 5 Ma better reflects the emplacement age of the Chaoyang diabase sills.

Twenty spots on 20 zircon grains from another diabase sample (09053-1) in Chaoyang were analyzed by LA-ICP-MS method. They yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1323 ± 11 Ma (95% confidence, MSWD = 0.33, N = 20; Fig. 8B), which is similar to the upper intercept age of the sample (1328 ± 10 Ma, MSWD = 0.54; the lower intercept age was forced through 0 Ma). High Th/U ratios (0.27–2.90), long euhedral prismatic morphology and weak oscillatory zoning (Fig. 7C) indicate that they are magmatic zircons. Therefore, the weighted mean $^{207}Pb/^{206}Pb$ age of 1323 ± 11 Ma is interpreted as the crystallization age of the diabase sills.

Although 16 baddeleyite grains were separated from sample 08100-1, only three of them are large enough for SIMS dating. Three analyses on three baddeleyite grains yield a weighed mean $^{207}Pb/^{206}Pb$ age of 1316 ± 37 Ma (95% confidence, MSWD = 1.6, Fig. 9B). This age is similar to that of baddeleyites from the diabase sills emplaced into the Wumishan Formation within errors, indicating their contemporaneous emplacement in the Mid-Mesoproterozoic.

Twenty-one analyses on 21 baddeleyite grains of sample SD057-4 from a diabase sill emplaced into the Xiamaling Formation in Pingquan were performed by SIMS and the results are plotted on a concordia diagram in Fig. 9C. Except for one analyses (3.1) with large errors, the remaining 20 analyses yield a weighed mean $^{207}Pb/^{206}Pb$ age of 1320 ± 4 Ma (95% confidence, MSWD = 1.2), which is interpreted as the emplacement age of the diabase sills within the Xiamaling Formation in Pingquan.

Fifteen spots on 15 baddeleyite grains on sample SD069-3 from a diabase sill emplaced into the Xiamaling Formation in Xiabancheng were analyzed by SIMS. They yield a weighed mean $^{207}Pb/^{206}Pb$ age of 1325 ± 5 Ma (95% confidence, MSWD = 1.3, N = 15, Fig. 9D), which reflects the emplacement age of the diabase sills within the Xiamaling Formation in Xiabancheng.

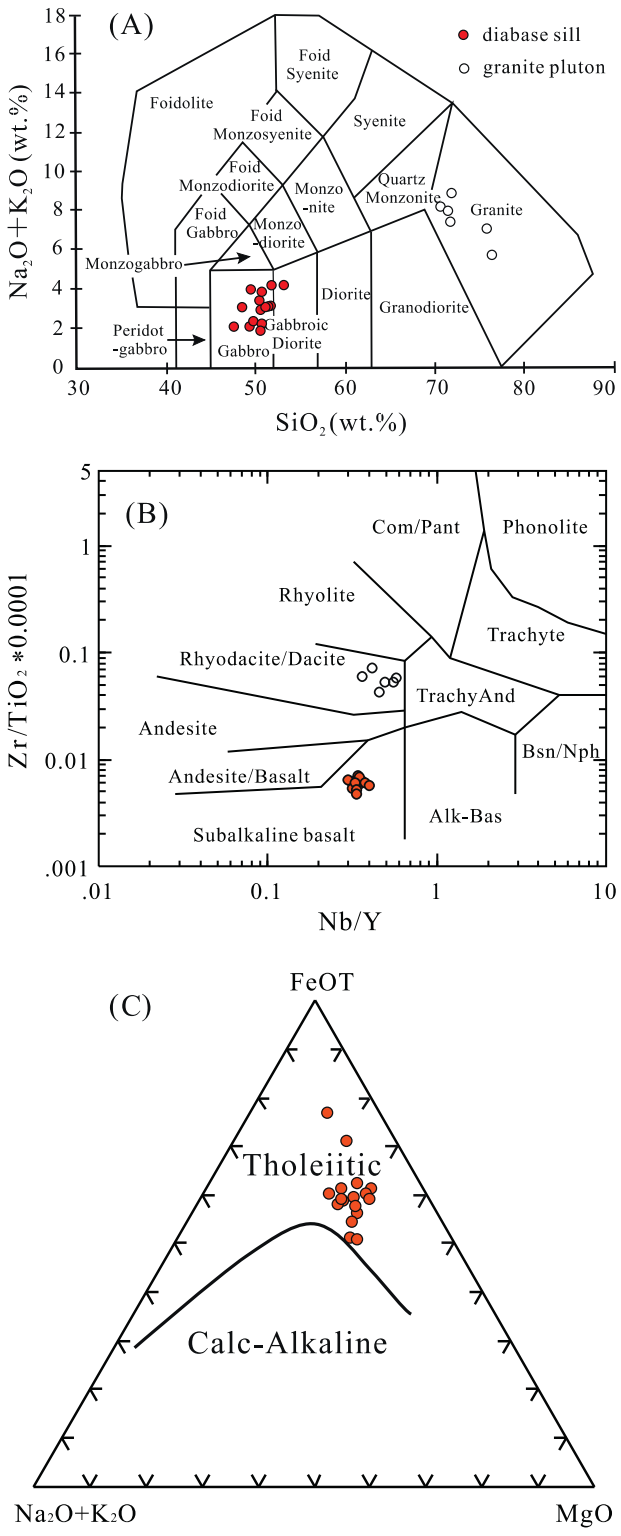


Fig. 10. Geochemical classification diagrams for Mid-Mesoproterozoic magmatic rocks from the northern part of the NCC. (A) Total alkali ($K_2O + Na_2O$) vs. silica (SiO_2) diagram (TAS) from Middlemost (1994); (B) Nb/Y vs. Zr/TiO_2 diagram from Winchester and Floyd (1977); (C) whole-rock alkali ($Na_2O + K_2O$)–FeOT–MgO (AFM) diagram from Irvine and Baragar (1971).

5.1.2. Granitic pluton

Four granite samples from the Shangdu-Huade pluton including three strongly deformed and one less deformed variety were chosen for zircon LA-ICP-MS U–Pb dating. The analytical results are listed in Table 2. Thirty spots on 29 zircon grains from a deformed biotite alkali feldspar granite sample (07014-2) in the central portion of the pluton were measured and plotted on a concordia diagram in Fig. 8C. Seven analyses on inherited zircons (02, 06, 09, 14, 15, 16, 24) yield relatively older $^{207}Pb/^{206}Pb$ ages ranging from 1503 ± 39 Ma to 1729 ± 37 Ma and reflect inherited ages from early magmatism or the country rocks. Except for the above inherited spots and one anomalous spot (11) caused by high common Pb contents, the remaining spots yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1331 ± 11 Ma (95% confidence, MSWD = 1.4, N = 22), which is similar to the upper intercept age of the sample (1341 ± 14 Ma, MSWD = 2.3; the lower intercept age was forced through 0 Ma). The long euhedral prismatic morphology and well-developed oscillatory zoning (Fig. 7F) indicate a magmatic origin for these zircons. Therefore, the weighted mean $^{207}Pb/^{206}Pb$ age of 1331 ± 11 Ma is considered to reflect the emplacement age of the deformed granite in the pluton.

Sixteen analyses were performed on 16 zircon grains from a less deformed biotite monzogranite sample (09316-2) in the southern part of the pluton. This sample includes a lot of inherited zircon grains or cores (01, 02, 04, 06, 12, 14, 16) yielding $^{207}Pb/^{206}Pb$ ages from 1529 ± 24 Ma to 1887 ± 22 Ma. The remaining spots yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1313 ± 17 Ma (95% confidence, MSWD = 0.37, N = 9, Fig. 8D), which is similar to the upper intercept age of the sample (1315 ± 15 Ma, MSWD = 0.61; the lower intercept age was forced through 0 Ma). Subhedral prismatic morphology and oscillatory zoning (Fig. 7G) indicate that they are magmatic in origin. Therefore, the weighted mean $^{207}Pb/^{206}Pb$ age of 1313 ± 17 Ma reflects the emplacement age of the less deformed granite in the pluton, which is slightly younger than that of the deformed granite.

Sixteen spots on 16 zircon grains from a deformed syenogranite sample (09319-2) in the central portion of the pluton were measured and plotted on a concordia diagram in Fig. 8E. Three inherited zircon grains (02, 06, 16) yield $^{207}Pb/^{206}Pb$ ages of 1488 ± 28 Ma to 1807 ± 24 Ma. The remaining 13 spots yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1324 ± 14 Ma (95% confidence, MSWD = 0.74), which is similar to the upper intercept age of the sample (1328 ± 12 Ma, MSWD = 1.09; the lower intercept age was forced through 0 Ma). Oscillatory zoning and subhedral prismatic morphology (Fig. 7H) indicate that these zircons are magmatic in origin. Therefore, the weighted mean $^{207}Pb/^{206}Pb$ age of 1324 ± 14 Ma is interpreted as the emplacement age of the deformed syenogranite.

Sixteen spots on 16 zircon grains from another deformed syenogranite sample (09323-1) in the northern part of the pluton were analyzed and plotted on a concordia diagram in Fig. 8F. Except for one older analyzed spot (05), the remaining analyses yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1330 ± 12 Ma (95% confidence, MSWD = 0.40, N = 15), which is similar to the upper intercept age of the sample (1334 ± 12 Ma, MSWD = 0.49; the lower intercept age was forced through 0 Ma). Strong oscillatory zoning and euhedral prismatic morphology in these zircons (Fig. 7I) suggests a magmatic origin. Therefore, the weighted mean $^{207}Pb/^{206}Pb$ age of 1330 ± 12 Ma represents the emplacement age of the deformed syenogranite.

5.2. Major and trace element compositions

The major and trace element data of the Mid-Mesoproterozoic magmatic rocks are listed in Table 3. The diabase sills are characterized by low contents of SiO_2 (47.08–51.03 wt.%), high contents of TiO_2 (1.51–3.72 wt.%), Fe_2O_3T (13.08–20.16 wt.%) and MgO

Table 3
Major and trace element compositions of the Mid-Mesoproterozoic magmatic rocks.

Number	SD057-1	SD057-2	SD057-3	SD057-4	SD069-1	SD069-2	SD069-3	08025-1	08029-2	09038-1	08093-1
Rock type	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase
Location	PQ	PQ	PQ	PQ	XBC	XBC	XBC	LY	LY	LY	CY
Longitude	118°45.5′	118°45.5′	118°45.5′	118°45.5′	118°07.9′	118°08.0′	118°08.0′	119°31.2′	119°33.2′	119°30.7′	120°18.4′
Latitude	40°58.8′	40°58.8′	40°58.6′	40°58.6′	40°41.3′	40°41.2′	40°41.2′	41°14.8	41°12.5′	41°13.3′	41°24.7′
Major element oxides (wt.%)											
SiO_2	49.73	49.29	50.05	49.35	50.20	51.03	47.08	50.02	48.68	50.13	48.51
TiO_2	2.53	2.52	2.09	2.22	1.93	2.08	3.10	1.96	1.93	1.80	3.72
Al_2O_3	14.35	14.45	14.59	13.97	13.72	14.79	11.61	13.34	14.31	13.40	11.91
Fe_2O_3T	15.28	15.46	14.54	14.93	14.33	13.43	20.02	14.75	15.24	13.73	20.16
MnO	0.17	0.24	0.19	0.18	0.23	0.18	0.23	0.19	0.19	0.19	0.23
MgO	5.97	7.45	5.16	5.05	6.63	5.17	5.12	6.32	6.67	6.11	3.26
CaO	1.35	1.46	7.80	7.69	9.58	8.79	9.55	10.42	8.47	10.86	8.12
Na_2O	2.98	3.12	1.80	1.96	1.66	1.90	1.67	1.30	1.13	2.14	1.15
K_2O	1.01	0.95	1.25	1.88	0.62	1.27	0.49	0.67	1.28	0.84	0.99
P_2O_5	0.24	0.24	0.18	0.20	0.16	0.17	0.23	0.17	0.18	0.16	0.31
LOI	5.99	4.90	2.43	2.12	1.20	1.38	0.72	1.00	1.92	0.64	1.46
Total	99.60	100.08	100.08	99.55	100.26	100.19	99.82	100.14	100.00	100.00	99.82
Mg#	43.63	48.85	41.29	40.13	47.84	43.28	33.64	45.92	46.45	46.87	24.27
Trace elements (ppm)											
La	18.14	16.85	14.89	15.78	11.60	14.01	15.91	13.09	13.16	12.76	22.32
Ce	35.91	38.58	32.82	35.24	25.65	30.78	35.87	29.05	29.41	28.94	50.56
Pr	5.07	5.00	4.20	4.51	3.32	3.90	4.72	3.81	3.94	3.70	6.67
Nd	22.39	22.26	18.42	20.65	15.62	17.37	22.20	18.19	17.03	17.61	30.82
Sm	5.30	5.52	4.33	5.17	3.81	4.20	5.53	4.64	4.25	4.28	7.73
Eu	1.24	1.46	1.68	1.74	1.19	1.44	1.66	1.50	1.36	1.49	2.44
Gd	5.89	5.67	4.35	5.48	4.05	4.57	6.14	5.04	4.70	4.75	8.05
Tb	0.79	0.82	0.67	0.76	0.55	0.59	0.85	0.66	0.66	0.71	1.10
Dy	5.62	6.11	4.59	6.04	4.17	4.50	6.58	5.04	4.85	5.25	8.74
Ho	1.28	1.30	1.00	1.31	0.88	0.93	1.39	1.11	1.04	1.11	1.88
Er	3.83	3.69	2.72	3.65	2.51	2.64	3.87	3.16	3.00	3.25	5.25
Tm	0.56	0.59	0.39	0.53	0.35	0.38	0.58	0.44	0.42	0.44	0.76
Yb	3.27	3.43	2.40	3.01	2.13	2.24	3.30	2.71	2.49	2.64	4.47
Lu	0.54	0.58	0.37	0.51	0.35	0.39	0.56	0.44	0.41	0.42	0.75
ΣREE	109.83	111.86	92.83	104.38	76.18	87.94	109.16	88.88	86.72	87.35	151.54
La_N/Yb_N	3.75	3.32	4.19	3.54	3.68	4.23	3.26	3.26	3.57	3.27	3.37
Eu_N/Eu_N^*	0.68	0.80	1.18	1.00	0.93	1.00	0.87	0.95	0.93	1.01	0.95
Li	62.36	37.96	19.72	28.00	10.10	11.20	11.82	10.18	18.19	10.08	19.09
Be	1.03	1.22	0.81	0.79	0.64	0.75	0.86	0.68	0.64	0.60	1.08
Sc	36.63	37.95	27.78	35.95	5.82	28.23	40.50	39.37	27.66	44.73	40.17
Cr	101.18	113.02	69.00	121.18	116.43	81.65	37.56	134.71	109.20	169.00	13.69
Co	49.49	44.97	41.19	41.23	41.52	34.35	49.35	44.00	47.38	41.53	45.55
V	427.33	447.35	357.89	422.39	359.11	346.77	847.74	420.02	320.12	429.66	633.62
Ni	83.74	86.59	68.17	56.45	80.74	38.42	58.32	67.43	126.92	64.39	28.02
Cu	75.53	78.74	17.51	63.59	56.36	26.69	74.60	57.89	62.51	51.07	119.30
Zn	83.13	93.14	69.97	122.50	78.19	72.84	88.48	58.25	70.61	73.21	109.66
Ga	20.36	22.71	20.33	20.44	18.20	19.24	21.13	19.61	18.38	18.89	24.45
Rb	16.14	17.00	40.27	53.82	16.21	31.12	13.84	19.68	35.58	23.53	31.44
Sr	52.14	56.71	167.95	244.70	149.30	160.40	135.50	155.50	160.20	170.60	196.40
Y	29.30	30.87	21.89	29.38	17.70	21.59	30.79	26.92	23.56	25.73	40.95
Zr	159.85	173.45	126.65	136.96	109.55	108.11	149.30	123.90	122.32	115.47	230.10
Nb	10.42	10.84	8.20	9.73	7.01	7.28	10.34	8.27	8.49	7.69	14.89
W	0.56	0.58	0.53	0.42	0.36	0.43	0.41	0.33	0.33	0.37	0.61
Mo	0.88	4.35	1.33	0.85	3.21	0.88	1.12	0.80	0.83	2.24	1.14
Sn	1.76	1.79	3.83	1.36	1.04	1.15	1.43	1.24	1.18	1.01	2.10
Cs	0.66	1.74	1.31	2.63	1.12	1.97	0.72	0.68	2.29	0.76	1.48
Ba	235.50	201.70	285.40	634.50	192.20	425.30	130.60	121.70	314.00	152.60	273.00
Hf	4.48	4.61	3.48	3.85	3.22	3.03	4.01	3.38	3.25	3.25	6.69
Ta	0.86	0.92	0.66	0.77	0.56	0.59	0.77	0.73	0.65	0.59	1.26
Pb	1.67	1.59	3.29	3.06	4.64	4.72	3.74	3.07	3.34	3.30	7.17
Th	3.36	2.98	3.00	2.41	1.18	2.92	2.51	2.17	2.03	2.13	3.58
U	0.78	0.67	0.64	0.48	0.56	0.62	0.54	0.47	0.46	0.48	0.72

Number	08100-1	09007-1	09008-1	09010-1	07014-2	09316-2	09318-1	09319-2	09319-3	09323-1
Rock type	Diabase	Diabase	Diabase	Diabase	Granite	Granite	Granite	Granite	Granite	Granite
Location	CY	KC	KC	KC	SH	SH	SH	SH	SH	SH
Longitude	120°14.2′	118°18.3′	118°17.5′	118°32.7′	113°46.9′	113°45.9′	113°46.0′	113°46.9′	113°46.8′	113°47.7′
Latitude	41°14.1′	40°36.6′	40°35.0′	40°39.8′	41°47.0′	41°44.6′	41°45.4′	41°47.0′	41°47.1′	41°48.6′
Major element oxides (wt.%)										
SiO_2	50.00	47.76	49.56	48.04	71.25	75.61	75.37	69.49	70.45	71.47
TiO_2	1.86	1.88	2.04	1.51	0.37	0.41	0.38	0.48	0.44	0.42
Al_2O_3	14.06	15.03	13.12	14.62	14.03	11.75	12.32	14.66	14.32	13.87
Fe_2O_3T	13.85	15.50	14.85	13.08	2.49	3.01	2.41	3.42	3.17	3.19
MnO	0.38	0.19	0.20	0.18	0.03	0.05	0.04	0.04	0.04	0.07
MgO	6.50	6.35	5.83	7.48	0.66	0.93	0.46	0.83	0.68	0.50

Table 3 (*Continued*)

Number	08100-1	09007-1	09008-1	09010-1	07014-2	09316-2	09318-1	09319-2	09319-3	09323-1
Major element oxides (wt.%)										
CaO	8.09	8.58	9.16	8.38	1.37	1.40	1.22	1.25	1.55	2.66
Na_2O	1.54	2.15	2.27	2.31	2.90	1.85	2.03	2.44	2.40	2.84
K_2O	1.55	0.93	1.21	1.64	5.99	3.88	5.04	5.69	5.53	4.61
P_2O_5	0.16	0.18	0.18	0.12	0.12	0.17	0.11	0.16	0.14	0.13
LOI	2.08	1.20	1.48	2.44	0.82	1.16	0.62	1.30	1.30	0.60
Total	100.07	99.75	99.90	99.80	100.03	100.22	100.00	99.76	100.02	100.36
Mg#	48.19	44.80	43.76	53.12	34.44	37.96	27.43	32.45	29.84	23.70
Trace elements (ppm)										
La	11.96	13.48	15.95	9.38	49.43	37.09	60.20	58.14	32.68	41.32
Ce	26.59	31.26	36.09	20.89	82.68	74.21	117.75	140.13	119.79	74.47
Pr	3.54	3.87	4.63	2.72	9.93	7.25	11.84	11.81	7.79	8.06
Nd	15.99	17.32	21.07	12.41	35.66	25.81	42.29	43.30	27.01	30.60
Sm	3.73	4.13	5.52	3.04	6.10	4.31	7.20	7.84	5.40	5.34
Eu	1.21	1.46	1.63	1.12	0.97	0.78	0.86	1.33	0.73	1.28
Gd	4.40	4.77	5.70	3.38	5.20	3.66	6.35	7.75	4.23	5.16
Tb	0.62	0.63	0.84	0.50	0.69	0.46	0.82	0.99	0.60	0.68
Dy	4.57	4.80	6.23	3.78	4.47	3.33	5.70	6.75	4.82	4.89
Ho	0.93	1.07	1.35	0.77	0.88	0.78	1.26	1.48	1.10	1.13
Er	2.68	3.07	3.87	2.31	2.54	2.41	3.56	4.25	3.33	3.34
Tm	0.39	0.43	0.55	0.34	0.35	0.36	0.52	0.62	0.49	0.50
Yb	2.32	2.55	3.33	1.92	1.95	2.12	3.11	3.54	3.14	3.12
Lu	0.41	0.45	0.58	0.34	0.34	0.35	0.49	0.61	0.51	0.51
ΣREE	79.34	89.29	107.34	62.90	201.19	162.92	261.95	288.54	211.62	180.40
La_N/Yb_N	3.48	3.57	3.24	3.30	17.13	11.82	13.08	11.10	7.03	8.95
Eu_N/Eu^*	0.91	1.01	0.89	1.07	0.53	0.60	0.39	0.52	0.47	0.75
Li	28.16	16.94	14.84	35.59	23.88	36.69	38.65	33.20	20.99	5.17
Be	0.72	0.64	1.01	0.55	1.95	3.01	2.20	2.87	2.26	1.35
Sc	34.30	27.35	41.95	28.58	6.02	5.74	4.46	9.46	7.12	5.98
Cr	121.49	83.13	101.97	139.70	17.63	23.05	9.96	30.96	18.50	26.81
Co	44.24	52.89	45.17	46.82	3.31	4.15	2.33	4.93	3.38	3.17
V	377.21	326.58	430.74	298.91	29.42	29.11	16.47	43.63	25.91	28.02
Ni	75.89	132.36	65.69	113.50	9.31	10.43	5.76	16.34	10.11	13.38
Cu	243.85	61.14	64.40	45.70	5.12	4.51	2.48	7.29	3.33	6.74
Zn	60.90	104.90	85.92	94.44	54.04	43.95	47.51	137.51	50.56	96.36
Ga	20.09	19.95	20.31	17.48	16.13	15.06	16.90	22.09	17.34	18.94
Rb	49.86	24.22	39.20	37.58	228.80	154.30	212.70	319.80	185.80	92.00
Sr	164.10	228.60	162.90	344.30	100.90	102.54	88.36	118.90	107.20	172.60
Y	21.12	24.57	28.58	17.89	21.85	19.55	31.32	41.44	26.85	28.43
Zr	109.91	116.09	145.31	80.07	195.25	174.10	227.60	257.40	256.70	306.10
Nb	7.35	8.46	9.79	5.60	12.13	8.95	11.24	20.51	15.38	11.68
W	0.41	0.48	0.54	0.34	1.45	1.56	1.29	2.01	0.61	0.69
Mo	0.76	3.28	2.91	3.78	0.68	1.31	0.60	2.14	0.75	2.71
Sn	1.28	1.12	1.28	0.85	2.06	2.24	2.05	4.74	2.06	1.79
Cs	3.46	1.96	0.95	1.49	7.50	6.64	5.92	19.35	5.16	1.34
Ba	366.50	317.60	206.90	575.00	750.40	433.70	540.60	816.00	635.60	1023.00
Hf	3.26	3.44	4.23	2.40	5.13	4.55	6.17	6.60	6.98	7.79
Ta	0.62	0.69	0.79	0.47	1.07	0.83	1.00	1.95	1.19	0.75
Pb	1.97	5.44	4.88	3.02	40.13	21.85	29.56	47.23	25.38	23.31
Th	2.25	2.13	2.68	1.52	14.72	12.51	22.64	22.97	20.26	10.55
U	0.47	0.55	0.67	0.37	1.35	2.37	1.85	2.46	2.03	1.79

Note: PQ – Pingquan; XBC – Xiabancheng; LY – Lingyuan; CY – Chaoyang; KC – Kuancheng; SH – Shangdu-Huade; Fe_2O_3T – Fe_2O_3 total; LOI – loss on ignition; Mg# = molecular MgO/(TFeO + MgO); Eu_N – chondrite-normalized Eu; $Eu^*_N = (Sm_N \times Gd_N)^{1/2}$.

(3.26–7.48 wt.%) and Mg# values from 24.27 to 53.12. Samples from the granitic pluton exhibit high contents of SiO_2 (69.49–75.61 wt.%), $K_2O + Na_2O$ (5.73–8.89 wt.%), low contents of TiO_2 (0.37–0.48 wt.%), Fe_2O_3T (2.41–3.42 wt.%) and MgO (0.46–0.93 wt.%) and low Mg# values of 23.70–37.96. According to the total alkali ($K_2O + Na_2O$) vs. silica (SiO_2) classification for plutonic rocks from Middlemost (1994), the rocks belong to gabbro-gabbroic diorite and granite, respectively (Fig. 10A). Plots on Nb/Y vs. Zr/TiO_2 diagram for volcanic rocks from Winchester and Floyd (1977), classify them as subalkaline basalt and rhyodacite/dacite, respectively (Fig. 10B). On whole-rock alkali ($Na_2O + K_2O$)–FeOT–MgO (AFM) diagram from Irvine and Baragar (1971), all the diabase samples exhibit tholeiitic compositions (Fig. 10C). The above results provide compelling indication that the Mid-Mesoproterozoic igneous rocks in northern NCC are bimodal in composition.

The samples from diabase sills emplaced into different layers within the Mesoproterozoic strata exhibit similar rare earth element (REE) compositions with total REE contents from 62.90 ppm to 151.54 ppm. Their chondrite-normalized REE patterns (Fig. 11A–E) are characterized by slight light REE (LREE)-enrichment (La_N/Yb_N = 3.24–4.19) and weak negative to no Eu anomalies (Eu_N/Eu_N = 0.68–1.18). Rocks from the Mid-Mesoproterozoic granite pluton are relatively rich in REE compositions with total REE contents from 162.92 ppm to 288.54 ppm. They exhibit significant LREE-enrichment (La_N/Yb_N = 7.03–17.13) and negative Eu anomalies (Eu_N/Eu_N = 0.39–0.75) on chondrite-normalized REE patterns (Fig. 11F).

Trace element compositions of the diabase sills emplaced into different layers within the Mesoproterozoic strata are also very similar and are characterized by relatively high contents of Cr, Co, V, Ni and Cu and low contents of Rb, Th and U (Table 3). On primitive mantle-normalized spidergrams (Fig. 12A–E), they display depletion in Th, U, Nb, Ta, Sr, P, Ti and Y and enrichment in K and Pb. The samples from the Shangdu-Huade granite pluton display strong

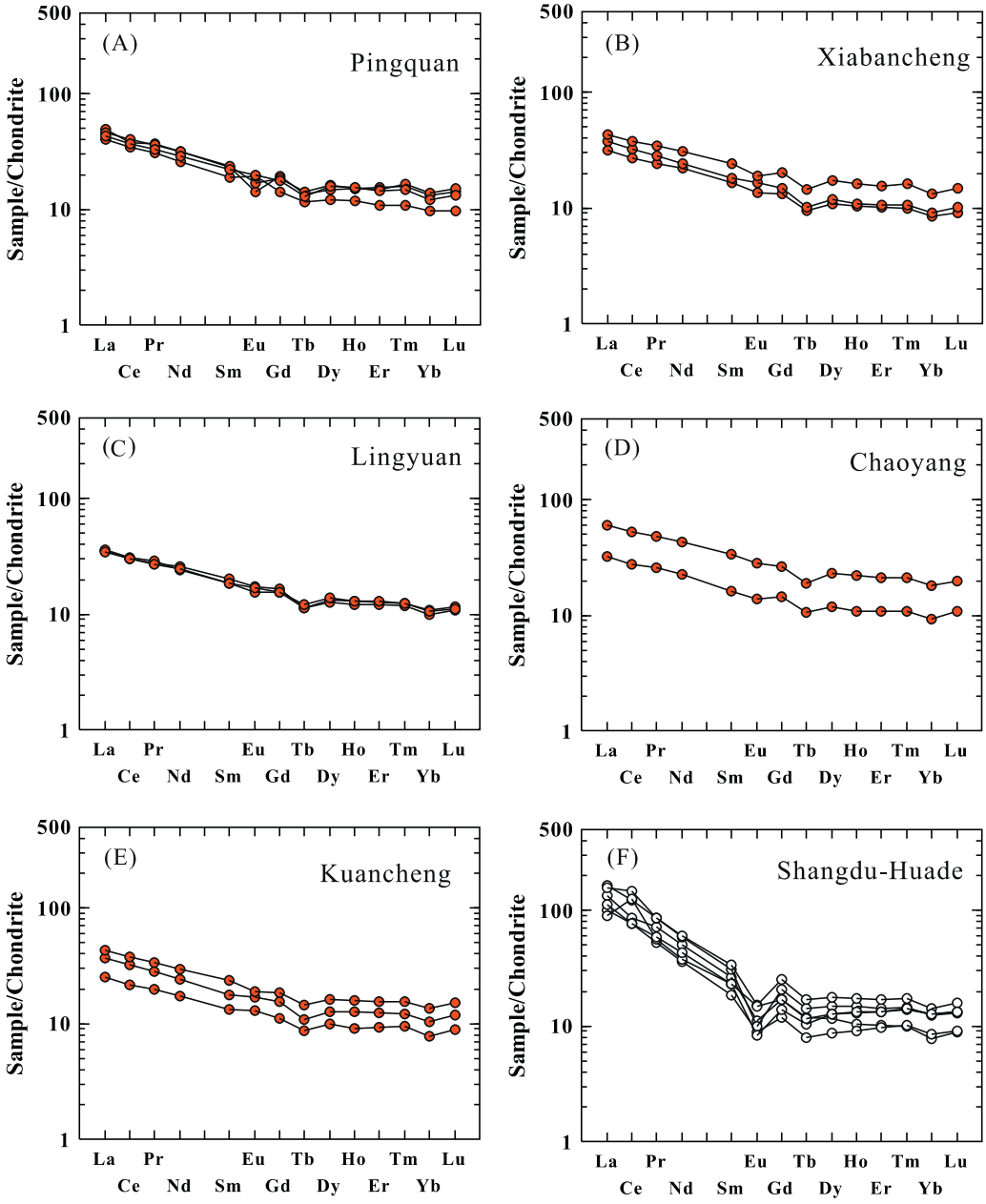


Fig. 11. Chondrite-normalized REE patterns for Mid-Mesoproterozoic magmatic rocks from the northern part of the NCC. The chondrite values are from Taylor and McLennan (1985).

depletion in Ba, U, Nb, Ta, Sr, P and Ti and enrichment in Rb, Th, K and Pb on primitive mantle-normalized spidergrams (Fig. 12F).

5.3. Sm–Nd isotopic data

Whole-rock Sm–Nd isotopic data of the Mid-Mesoproterozoic diabase and granite rocks are listed in Table 4 and plotted in Fig. 13A. All the diabase samples from the Wumishan, Tieling and Xiamaling Formations in different locations (Pingquan, Xiabancheng, Lingyuan, Chaoyang and Kuancheng) exhibit very similar Nd isotopic compositions with initial $^{143}Nd/^{144}Nd$ ratios from 0.510933 to 0.511026, positive $\varepsilon_{Nd}(t)$ values from 0.14 to 1.96, and Nd isotopic T_{DM} model ages from 1.90 Ga to 2.18 Ga. The granite samples from the Shangdu-Huade pluton are characterized by low initial $^{143}Nd/^{144}Nd$ ratios from 0.510565 to 0.510595, low negative $\varepsilon_{Nd}(t)$ values from −6.94 to −6.35, and Nd isotopic T_{DM} model ages from 2.28 Ga to 2.37 Ga. The significant difference in whole-rock Nd isotopic compositions of the diabase sills and granite pluton indicates a different magma source for the diabasic and granitic rocks in the northern NCC.

5.4. Lu–Hf isotopes in zircon and baddeleyite

Lu–Hf isotopic compositions in zircons and baddeleyites from the Mid-Mesoproterozoic magmatic rocks are listed in Table 5 and plotted in Fig. 13B. There are significant differences in Hf isotopic compositions between the diabase and granite analyzed in this study. Zircons from the diabase samples (08093-1, 09053-1) are characterized by high initial $^{176}Hf/^{177}Hf$ ratios from 0.282023 to 0.282179, high positive $\varepsilon_{Hf}(t)$ values from 2.93 to 8.48, and young Hf isotopic T_{DM} model ages from 1.49 Ga to 1.75 Ga and T_{DM}^{C} from 1.57 Ga to 1.93 Ga. Baddeleyites from the diabase sample (08093-1) display similar Hf isotopic compositions with high initial $^{176}Hf/^{177}Hf$ ratios from 0.281997 to 0.282085, high positive

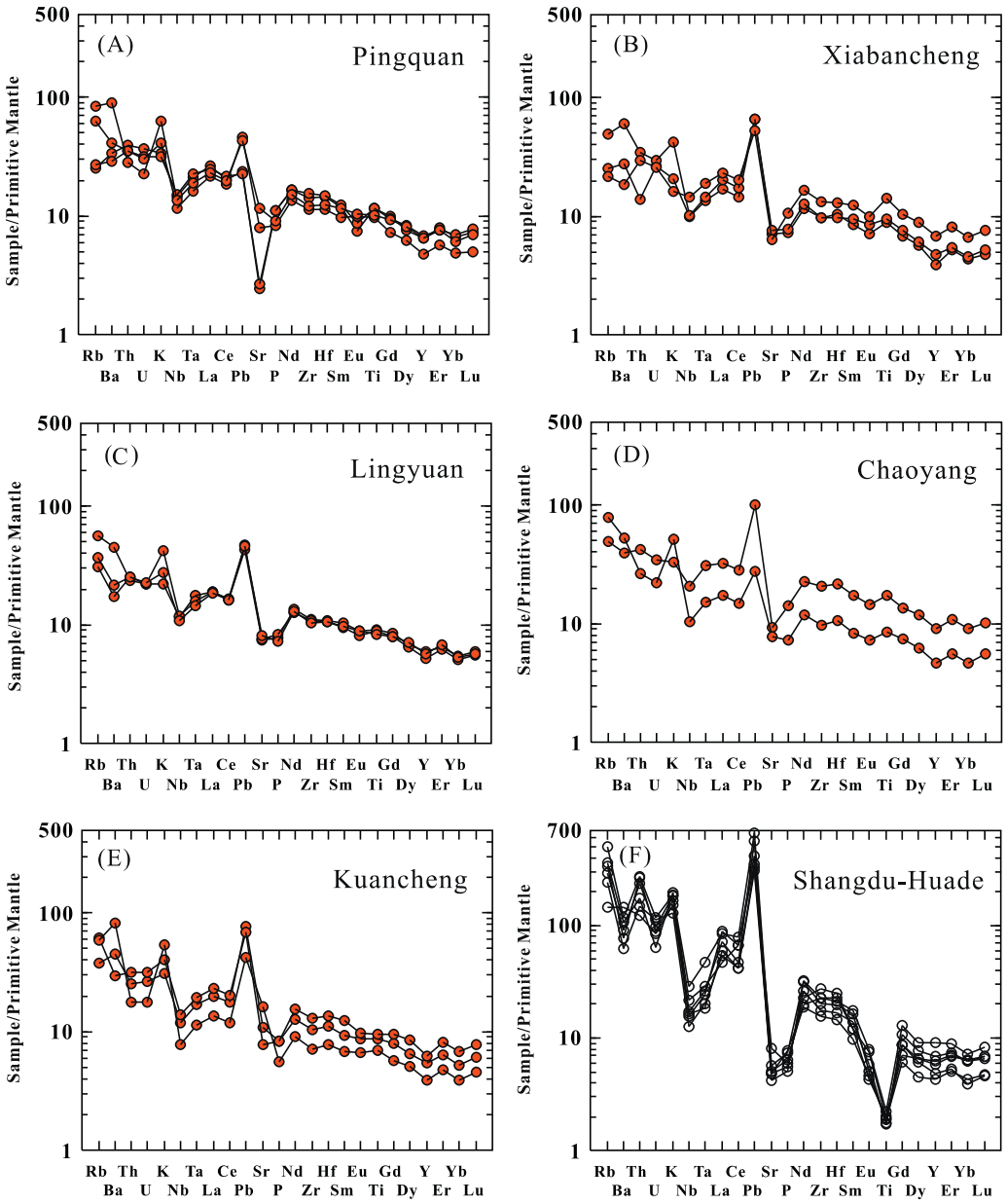


Fig. 12. Primitive mantle-normalized spidergrams for Mid-Mesoproterozoic magmatic rocks from the northern part of the NCC. The primitive mantle values are from Sun and McDonough (1989).

$\varepsilon_{Hf}(t)$ values from 2.00 to 5.13, and young Hf isotopic T_{DM} model ages from 1.60 Ga to 1.73 Ga and T_{DM}^{C} from 1.79 Ga to 1.99 Ga.

However, syn-magmatic zircons from the Shangdu-Huade granite samples display different Hf isotopic compositions and are characterized by low initial $^{176}Hf/^{177}Hf$ ratios, negative $\varepsilon_{Hf}(t)$ values and old Hf isotopic T_{DM} and T_{DM}^{C} model ages. The syn-magmatic zircons from the deformed granite sample (09323-1) in the northern part of the pluton exhibit low initial $^{176}Hf/^{177}Hf$ ratios from 0.281690 to 0.281804, negative $\varepsilon_{Hf}(t)$ values from −8.76 to −4.73, and old Hf isotopic T_{DM} model ages from 2.00 Ga to 2.16 Ga and T_{DM}^{C} from 2.42 Ga to 2.67 Ga. Most of the syn-magmatic zircons from the deformed granite samples (07014-2, 09319-2) in the central portion of the pluton are characterized by low initial $^{176}Hf/^{177}Hf$ ratios from 0.281639 to 0.281886, negative $\varepsilon_{Hf}(t)$ values from −10.57 to −1.80, and old Hf isotopic T_{DM} model ages from 1.88 Ga to 2.24 Ga and T_{DM}^{C} from 2.23 Ga to 2.79 Ga. Most of the syn-magmatic zircons from the less deformed granite sample (09316-2) in the southern part of the pluton display low initial $^{176}Hf/^{177}Hf$ ratios from 0.281635 to 0.281818, negative $\varepsilon_{Hf}(t)$ values from −11.05 to −4.56, and old Hf isotopic T_{DM} model ages from 2.00 Ga to 2.22 Ga and T_{DM}^{C} from 2.40 Ga to 2.81 Ga. Very low initial $^{176}Hf/^{177}Hf$ ratios of two spots (spot 11 in sample 07014-2 and spot 10 in sample 09316-2) probably resulted from influence of old inherited cores within the analyzed zircon grains.

6. Discussion

6.1. Ages of bimodal magmatism and comparison with Mid-Mesoproterozoic (1.4–1.2 Ga) magmatism in other cratons

Although the occurrence of a large volume of diabase sills within the late Paleoproterozoic to Mesoproterozoic sedimentary rocks in the northern NCC was known since several decades, their emplacement ages remained controversial largely due to the lack of reliable geochronological data (LBGMR, 1965, 1967, 1969; HBGMR, 1969, 1972; Li and Li, 2005). Baddeleyite (ZrO_2), which is a common

Table 4
Sm–Nd isotopes of the Mid-Mesoproterozoic magmatic rocks.

Sample no.	Sm	Nd	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$ (2σ)	$(^{143}Nd/^{144}Nd)_i$	$\varepsilon_{Nd}(0)$	$\varepsilon_{Nd}(t)$	$f_{Sm/Nd}$	T_{DM} (Ma)
Diabase sills (age = 1325 Ma)									
SD057-2	6.34	26.39	0.1455	0.512201 ± 14	0.510935	−8.52	0.17	−0.26	2112
SD057-4	5.67	22.68	0.1513	0.512278 ± 10	0.510961	−7.02	0.69	−0.23	2123
SD069-3	6.21	24.86	0.1512	0.512284 ± 12	0.510967	−6.91	0.81	−0.23	2106
08025-1	5.08	20.10	0.1530	0.512310 ± 10	0.510979	−6.39	1.03	−0.22	2101
08029-2	5.15	20.69	0.1508	0.512263 ± 11	0.510950	−7.32	0.48	−0.23	2141
08093-1	7.93	32.17	0.1492	0.512286 ± 14	0.510987	−6.87	1.20	−0.24	2035
08100-1	4.43	17.81	0.1505	0.512243 ± 15	0.510933	−7.71	0.14	−0.23	2180
09007-1	5.00	20.08	0.1507	0.512258 ± 12	0.510946	−7.41	0.40	−0.23	2150
*CY-3	3.73	14.84	0.1519	0.512343 ± 8	0.511021	−5.75	1.86	−0.23	1984
*CY-5	3.62	14.92	0.1467	0.512238 ± 9	0.510961	−7.80	0.69	−0.25	2067
*CY-7	3.93	16.48	0.1443	0.512282 ± 9	0.511026	−6.94	1.96	−0.27	1901
*CY-9	3.39	13.88	0.1477	0.512234 ± 9	0.510949	−7.88	0.44	−0.25	2108
*CY-15	3.31	13.78	0.1450	0.512242 ± 9	0.510980	−7.72	1.06	−0.26	2008
*CY-16	3.84	16.23	0.1430	0.512208 ± 8	0.510963	−8.39	0.73	−0.27	2024
Granite pluton (age = 1330 Ma)									
07014-2	7.42	42.87	0.1047	0.511510 ± 12	0.510595	−22.00	−6.35	−0.47	2284
09323-1	6.25	34.57	0.1094	0.511520 ± 13	0.510565	−21.80	−6.94	−0.44	2370

$T_{DM} = 1/\lambda \times \ln\{1 + [(^{143}Nd/^{144}Nd)_{sample} - (^{143}Nd/^{144}Nd)_{DM}]/[(^{147}Sm/^{144}Nd)_{sample} - (^{147}Sm/^{144}Nd)_{DM}]\}$; $f_{Sm/Nd} = (^{147}Sm/^{144}Nd)_{sample}/(^{147}Sm/^{144}Nd)_{CHUR} - 1$. The $^{143}Nd/^{144}Nd$ and $^{147}Sm/^{144}Nd$ ratios of chondrite and depleted mantle at present day are 0.512638 and 0.1967, 0.51315 and 0.2137, respectively (Jacobsen and Wasserburg, 1980). Samples marked with asterisk are recalculated from Li and Li (2005).

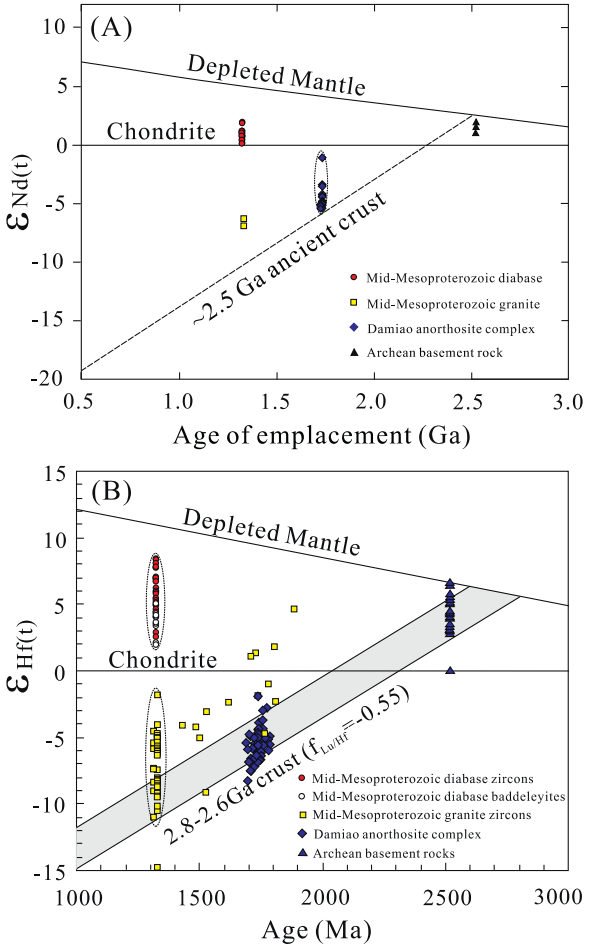


Fig. 13. (A) $\varepsilon_{Nd}(t)$ vs. emplacement age and (B) $\varepsilon_{Hf}(t)$ vs. U–Pb age of zircons and baddeleyites from Mid-Mesoproterozoic magmatic rocks from the northern part of the NCC. The ancient crust evolution line in (A) is constructed on the basis of an average $^{147}Sm/^{144}Nd$ value of 0.118 (Jahn and Condie, 1995). Nd–Hf isotopic data of the Damiao anorthosite complex are from Xie and Wang (1988), Zhang et al. (2007a) and Zhao et al. (2009a), and those of the Archean basement rocks are from Ramo et al. (1995) and Yang et al. (2005).

accessory mineral in some mafic-ultramafic or alkaline rocks, provides a powerful tool to determine the emplacement ages of such rocks (e.g., Krogh et al., 1987; Kamo et al., 1989; Heaman and LeCheminant, 1993; Wingate et al., 1998; Söderlund and Johansson, 2002; Li et al., 2007; Heaman, 2009). In some cases, syn-magmatic zircons can be obtained from silica-unsaturated rocks such as diabase and gabbro for high precision in situ U–Pb analysis (e.g., Kamo et al., 1989; Wingate et al., 1998; Holm et al., 2006; Zhu et al., 2008). Therefore, in this study, both zircons and baddeleyite were dated to determine the emplacement ages of the diabase sills. Although some of the recent zircon and baddeleyite U–Pb data (Zhang et al., 2009b; Li et al., 2009a) suggest that emplacement of these diabase sills occurred during Mid-Mesoproterozoic time, the ages previously obtained from the sills within the Wumishan Formation (ca. 1.35 Ga) are slightly older than those within the Xiamaling Formation (ca. 1.32 Ga). Therefore, whether the diabase sills within different formations in the late Paleoproterozoic to Mesoproterozoic sedimentary rocks in the northern NCC were emplaced simultaneously is still uncertain and more data are essential to confirm the emplacement ages of the diabase sills.

Our new SHRIMP and LA-ICP-MS U–Pb zircon ages from two diabase samples (Fig. 8A–B) and SIMS baddeleyite dating of four diabase samples (Fig. 9A–D) emplaced into the Wumishan, Tieling and Xiamaling Formations yield similar emplacement ages ranging from 1325 ± 5 Ma to 1316 ± 37 Ma. These results indicate that the emplacement of the diabase sills within Wumishan, Tieling and Xiamaling Formations occurred simultaneously at around 1.33–1.32 Ga. The ages are consistent with the field occurrence of the diabase sills. The uppermost stratum into which the diabase sills were emplaced is the Xiamaling Formation, and no diabase sills were found within the Changlongshan and Jingeryu Formations conformably covering the Xiamaling Formation (Fig. 3). This observation suggests that emplacement of these diabase sills occurred between the time of deposition of the Xiamaling and Changlongshan Formations. We thus infer that emplacement of large volumes of diabase sills (named as the Yanliao mafic sill swarms) into the late Paleoproterozoic to Mesoproterozoic sedimentary in the northern NCC must have occurred in the Mid-Mesoproterozoic at around 1.33–1.32 Ga. These emplacement ages are slightly younger than the depositional ages of the Xiamaling Formation constrained by the 1366 ± 9 Ma to 1380 ± 36 Ma ages from K-bentonite beds (Gao et al., 2007, 2008b; Su et al., 2008, 2010) and are consistent with the

Table 5
Hf isotope analyses results of zircons and baddeleyites from the Mid-Mesoproterozoic magmatic rocks.

Spots	Age (Ma)	$^{176}Yb/^{177}Hf$	$^{176}Lu/^{177}Hf$	$^{176}Hf/^{177}Hf$	$\pm2\sigma$	$\varepsilon_{Hf}(0)$	$\varepsilon_{Hf}(T)$	T_{DM} (Ma)	T^{C}_{DM} (Ma)	Hf_i	$f_{Lu/Hf}$
Diabase sample 08093-1 (zircon)											
01	**1325**	0.244998	0.004179	0.282201	0.000021	−20.20	5.52	1618	1763	0.282096	−0.87
02	**1325**	0.214162	0.003956	0.282160	0.000021	−21.64	4.28	1668	1842	0.282061	−0.88
03	**1325**	0.482607	0.008497	0.282304	0.000031	−16.55	5.34	1668	1774	0.282091	−0.74
04	**1325**	0.192590	0.003197	0.282190	0.000019	−20.59	6.00	1589	1732	0.282110	−0.90
05	**1325**	0.310481	0.005454	0.282217	0.000044	−19.63	4.96	1654	1799	0.282080	−0.84
06	**1325**	0.241585	0.004374	0.282288	0.000022	−17.12	8.43	1494	1576	0.282178	−0.87
07	**1325**	0.212302	0.004323	0.282151	0.000021	−21.98	3.61	1701	1885	0.282042	−0.87
08	**1325**	0.197559	0.003672	0.282133	0.000027	−22.59	3.57	1695	1888	0.282041	−0.89
09	**1325**	0.146293	0.002625	0.282123	0.000019	−22.96	4.14	1661	1851	0.282057	−0.92
10	**1325**	0.159630	0.002839	0.282177	0.000021	−21.03	5.88	1592	1740	0.282106	−0.91
11	**1325**	0.169119	0.003451	0.282194	0.000026	−20.44	5.93	1594	1737	0.282108	−0.90
12	**1325**	0.357504	0.006899	0.282239	0.000048	−18.86	4.45	1692	1831	0.282066	−0.79
13	**1325**	0.204609	0.003862	0.282168	0.000030	−21.35	4.65	1652	1819	0.282072	−0.88
14	**1325**	0.333623	0.006149	0.282177	0.000034	−21.04	2.93	1753	1928	0.282023	−0.81
15	**1325**	0.190131	0.003397	0.282190	0.000021	−20.58	5.84	1598	1743	0.282105	−0.90
16	**1325**	0.334591	0.006723	0.282228	0.000044	−19.25	4.21	1701	1847	0.282059	−0.80
17	**1325**	0.209848	0.003629	0.282229	0.000024	−19.21	7.01	1550	1668	0.282138	−0.89
Diabase sample 08093-1 (baddeleyite)											
01	**1325**	0.030293	0.000543	0.282099	0.000024	−23.81	5.13	1604	1788	0.282085	−0.98
02	**1325**	0.050299	0.000844	0.282022	0.000024	−26.53	2.14	1724	1979	0.282001	−0.97
03	**1325**	0.039225	0.000593	0.282059	0.000023	−25.22	3.67	1661	1881	0.282044	−0.98
04	**1325**	0.028922	0.000495	0.282074	0.000038	−24.70	4.28	1637	1842	0.282061	−0.99
05	**1325**	0.042128	0.000609	0.282057	0.000042	−25.28	3.60	1665	1886	0.282042	−0.98
06	**1325**	0.020953	0.000362	0.282045	0.000025	−25.70	3.39	1670	1899	0.282036	−0.99
07	**1325**	0.038339	0.000789	0.282017	0.000030	−26.71	2.00	1728	1988	0.281997	−0.98
08	**1325**	0.020997	0.000370	0.282073	0.000038	−24.74	4.36	1633	1837	0.282063	−0.99
09	**1325**	0.020485	0.000426	0.282070	0.000040	−24.84	4.20	1640	1847	0.282059	−0.99
10	**1325**	0.045413	0.001023	0.282027	0.000043	−26.35	2.16	1725	1978	0.282001	−0.97
11	**1325**	0.033074	0.000718	0.282032	0.000028	−26.18	2.60	1704	1950	0.282014	−0.98
Diabase sample 09053-1 (zircon)											
01	**1325**	0.118682	0.001581	0.282157	0.000016	−21.74	6.28	1567	1714	0.282118	−0.95
02	**1325**	0.235935	0.004438	0.282166	0.000047	−21.42	4.07	1682	1856	0.282055	−0.87
03	**1325**	0.241901	0.004992	0.282236	0.000021	−18.96	6.05	1602	1729	0.282111	−0.85
04	**1325**	0.243407	0.004642	0.282207	0.000031	−19.99	5.32	1630	1776	0.282091	−0.86
05	**1325**	0.100304	0.001533	0.282124	0.000015	−22.91	5.15	1611	1787	0.282086	−0.95
06	**1325**	0.203839	0.003061	0.282256	0.000018	−18.24	8.48	1486	1573	0.282179	−0.91
07	**1325**	0.182575	0.002864	0.282233	0.000034	−19.06	7.83	1511	1615	0.282161	−0.91
08	**1325**	0.216454	0.003920	0.282179	0.000021	−20.98	4.97	1639	1798	0.282081	−0.88
09	**1325**	0.164786	0.002821	0.282115	0.000027	−23.23	3.68	1682	1880	0.282044	−0.92
10	**1325**	0.174960	0.003408	0.282225	0.000022	−19.34	7.07	1546	1664	0.282140	−0.90
11	**1325**	0.314421	0.005963	0.282245	0.000032	−18.62	5.52	1634	1763	0.282096	−0.82
12	**1325**	0.263201	0.004391	0.282280	0.000023	−17.41	8.13	1507	1596	0.282170	−0.87
13	**1325**	0.159454	0.002699	0.282149	0.000018	−22.03	5.00	1627	1796	0.282081	−0.92
14	**1325**	0.225076	0.003342	0.282214	0.000025	−19.74	6.73	1560	1685	0.282130	−0.90
15	**1325**	0.221961	0.003594	0.282227	0.000029	−19.27	6.97	1551	1670	0.282137	−0.89
16	**1325**	0.166692	0.002619	0.282226	0.000015	−19.31	7.80	1511	1617	0.282160	−0.92
Granite sample 07014-2 (zircon)											
01	**1330**	0.033828	0.001085	0.281696	0.000020	−38.06	−9.53	2186	2722	0.281669	−0.97
02	1503	0.020611	0.000506	0.281698	0.000016	−37.97	−5.08	2149	2575	0.281684	−0.98
03	**1330**	0.042963	0.001502	0.281677	0.000034	−38.72	−10.57	2236	2787	0.281639	−0.95
04	**1330**	0.027054	0.000758	0.281729	0.000015	−36.90	−8.08	2122	2630	0.281710	−0.98
05	**1330**	0.069773	0.002458	0.281758	0.000033	−35.85	−8.54	2178	2659	0.281697	−0.93
06	1711	0.023147	0.000618	0.281746	0.000018	−36.29	1.11	2091	2343	0.281726	−0.98
07	**1330**	0.031868	0.000900	0.281693	0.000016	−38.17	−9.48	2180	2719	0.281670	−0.97
08	**1330**	0.046274	0.001404	0.281712	0.000019	−37.50	−9.25	2182	2704	0.281676	−0.96
09	1729	0.042844	0.001535	0.281771	0.000024	−35.39	1.36	2106	2342	0.281721	−0.95
10	**1330**	0.042463	0.001230	0.281724	0.000016	−37.06	−8.66	2155	2667	0.281693	−0.96
11	**1330**	0.042441	0.001342	0.281553	0.000019	−43.09	−14.81	2398	3053	0.281520	−0.96
12	**1330**	0.038895	0.001290	0.281737	0.000019	−36.61	−8.26	2141	2642	0.281704	−0.96
13	**1330**	0.049200	0.001728	0.281692	0.000024	−38.18	−10.23	2228	2765	0.281649	−0.95
14	**1330**	0.036918	0.001184	0.281756	0.000027	−35.92	−7.48	2108	2592	0.281726	−0.96
15	**1330**	0.067897	0.001870	0.281753	0.000017	−36.02	−8.19	2151	2637	0.281706	−0.94
16	**1330**	0.032336	0.001051	0.281913	0.000022	−30.38	−1.80	1884	2234	0.281886	−0.97
17	**1330**	0.057103	0.001822	0.281717	0.000022	−37.33	−9.45	2200	2717	0.281671	−0.95
18	**1330**	0.031660	0.000911	0.281671	0.000017	−38.95	−10.27	2210	2768	0.281648	−0.97
Granite sample 09323-1 (zircon)											
01	**1330**	0.069921	0.001611	0.281745	0.000016	−36.33	−8.27	2148	2643	0.281704	−0.95
02	**1330**	0.056899	0.001060	0.281785	0.000017	−34.92	−6.36	2062	2522	0.281758	−0.97
03	**1330**	0.046449	0.000874	0.281791	0.000016	−34.70	−5.97	2043	2498	0.281769	−0.97
04	**1330**	0.042581	0.000810	0.281782	0.000019	−35.00	−6.22	2052	2513	0.281762	−0.98
05	1433	0.052683	0.001238	0.281789	0.000024	−34.75	−4.11	2065	2460	0.281756	−0.96
06	**1330**	0.038464	0.000771	0.281813	0.000019	−33.90	−5.08	2007	2441	0.281794	−0.98
07	**1330**	0.034410	0.000750	0.281728	0.000019	−36.94	−8.11	2123	2632	0.281709	−0.98

Table 5 (*Continued*)

Spots	Age (Ma)	$^{176}Yb/^{177}Hf$	$^{176}Lu/^{177}Hf$	$^{176}Hf/^{177}Hf$	$\pm2\sigma$	$\varepsilon_{Hf}(0)$	$\varepsilon_{Hf}(T)$	T_{DM} (Ma)	T_{DM}^{C} (Ma)	Hf_i	$f_{Lu/Hf}$
Granite sample 09323-1 (zircon)											
08	**1330**	0.066664	0.001385	0.281808	0.000022	−34.08	−5.81	2047	2487	0.281773	−0.96
09	**1330**	0.078546	0.001807	0.281807	0.000019	−34.14	−6.25	2072	2515	0.281761	−0.95
10	**1330**	0.050647	0.001155	0.281723	0.000023	−37.08	−8.62	2152	2664	0.281694	−0.97
11	**1330**	0.045567	0.000985	0.281829	0.000023	−33.36	−4.73	1997	2419	0.281804	−0.97
12	**1330**	0.056396	0.001181	0.281720	0.000019	−37.20	−8.76	2158	2673	0.281690	−0.96
13	**1330**	0.077726	0.001676	0.281800	0.000016	−34.39	−6.38	2075	2523	0.281757	−0.95
14	**1330**	0.036480	0.000712	0.281783	0.000018	−34.98	−6.11	2045	2506	0.281765	−0.98
15	**1330**	0.101917	0.002020	0.281844	0.000017	−32.82	−5.11	2031	2443	0.281793	−0.94
Granite sample 09319-2 (zircon)											
01	**1330**	0.038572	0.000708	0.281816	0.000014	−33.81	−4.93	2000	2432	0.281798	−0.98
02	1807	0.078243	0.001825	0.281746	0.000014	−36.27	1.81	2158	2373	0.281684	−0.95
03	**1330**	0.080149	0.001444	0.281812	0.000013	−33.95	−5.73	2045	2482	0.281776	−0.96
04	**1330**	0.093506	0.002024	0.281844	0.000017	−32.80	−5.10	2031	2442	0.281793	−0.94
05	**1330**	0.037008	0.000691	0.281697	0.000014	−38.00	−9.12	2161	2696	0.281680	−0.98
06	1783	0.049318	0.000900	0.281650	0.000012	−39.66	−1.00	2237	2534	0.281620	−0.97
07	**1330**	0.077523	0.001537	0.281825	0.000015	−33.50	−5.36	2032	2459	0.281786	−0.95
08	**1330**	0.052253	0.000971	0.281802	0.000013	−34.32	−5.68	2034	2479	0.281777	−0.97
09	**1330**	0.097577	0.002117	0.281844	0.000012	−32.81	−5.19	2036	2448	0.281791	−0.94
10	**1330**	0.083372	0.001801	0.281844	0.000017	−32.82	−4.92	2019	2431	0.281799	−0.95
11	**1330**	0.061767	0.001151	0.281832	0.000014	−33.23	−4.75	2001	2420	0.281803	−0.97
12	**1330**	0.070020	0.001342	0.281838	0.000012	−33.01	−4.70	2002	2417	0.281805	−0.96
13	**1330**	0.078517	0.001555	0.281835	0.000015	−33.14	−5.02	2019	2437	0.281796	−0.95
14	**1330**	0.068098	0.001428	0.281859	0.000015	−32.28	−4.05	1978	2376	0.281823	−0.96
15	**1330**	0.127886	0.002985	0.281860	0.000016	−32.24	−5.39	2061	2461	0.281785	−0.91
16	1488	0.073656	0.001609	0.281763	0.000016	−35.68	−4.23	2122	2510	0.281718	−0.95
Granite sample 09316-2 (zircon)											
01	1621	0.059899	0.001125	0.281720	0.000013	−37.20	−2.35	2154	2494	0.281686	−0.97
02	1532	0.061377	0.001240	0.281758	0.000014	−35.87	−3.08	2109	2471	0.281722	−0.96
03	**1315**	0.095631	0.002064	0.281832	0.000016	−33.23	−5.88	2050	2480	0.281781	−0.94
04	1809	0.065975	0.001490	0.281617	0.000018	−40.83	−2.32	2318	2637	0.281566	−0.96
05	**1315**	0.082893	0.001964	0.281759	0.000016	−35.82	−8.39	2148	2638	0.281710	−0.94
06	1529	0.066733	0.001569	0.281598	0.000014	−41.52	−9.16	2350	2853	0.281552	−0.95
07	**1315**	0.047291	0.000914	0.281761	0.000014	−35.76	−7.40	2087	2576	0.281738	−0.97
08	**1315**	0.052104	0.001051	0.281718	0.000012	−37.27	−9.04	2153	2679	0.281692	−0.97
09	**1315**	0.055153	0.001016	0.281764	0.000015	−35.66	−7.39	2088	2575	0.281738	−0.97
10	**1315**	0.047055	0.000937	0.281523	0.000014	−44.19	−15.87	2415	3107	0.281499	−0.97
11	**1315**	0.104207	0.002344	0.281876	0.000016	−31.67	−4.56	2003	2397	0.281818	−0.93
12	1739	0.108764	0.002689	0.281713	0.000016	−37.46	−1.86	2258	2554	0.281624	−0.92
13	**1315**	0.036216	0.000695	0.281653	0.000015	−39.59	−11.05	2223	2806	0.281635	−0.98
14	1766	0.056382	0.001130	0.281563	0.000013	−42.74	−4.74	2371	2758	0.281526	−0.97
15	**1315**	0.073895	0.001476	0.281830	0.000013	−33.32	−5.45	2021	2453	0.281793	−0.96
16	1887	0.043245	0.000953	0.281747	0.000014	−36.26	4.65	2108	2253	0.281712	−0.97

^{176}Lu decay constant $\lambda = 1.867 \times 10^{-11}$ yr^{-1} (Söderlund et al., 2004); chondritic values: $^{176}Lu/^{177}Hf$ (0.0332 ± 0.0002, $^{176}Hf/^{177}Hf$ (0.282772 ± 0.000029; depleted mantle values: $(^{176}Lu/^{177}Hf)_{DM}$ (0.0384, $(^{176}Hf/^{177}Hf)_{DM}$ (0.28325; Hf_i: initial Hf isotope composition for U-Pb age; $T_{DM} = 1/\lambda \times \ln\{1 + [(^{176}Hf/^{177}Hf)_{sample} - (^{176}Hf/^{177}Hf)_{DM}]/[(^{176}Lu/^{177}Hf)_{sample} - (^{176}Lu/^{177}Hf)_{DM}]\}$; $T_{DM}^{C} = 1/\lambda \times \ln\left\{1 + \left[\left(^{176}Hf/^{177}Hf\right)_{sample,t} - \left(^{176}Hf/^{177}Hf\right)_{DM,t}\right] / \left[\left(^{176}Lu/^{177}Hf\right)_{c} - \left(^{176}Lu/^{177}Hf\right)_{DM}\right]\right\} + t$; $(^{176}Lu/^{177}Hf)_c = 0.015$; t = crystallization time of zircon.

fact that emplacement of the diabase sills occurred after deposition of the Xiamling Formation.

Zircon LA-ICP-MS U–Pb dating on four samples from the Shangdu-Huade granite pluton in the northern NCC yielded emplacement ages from 1331 ± 11 Ma to 1313 ± 17 Ma (Fig. 9C–F). These ages are similar to those obtained from the diabase sills in the northern NCC, indicating coeval emplacement of these bimodal intrusive rocks in the Mid-Mesoproterozoic at around 1.33–1.31 Ga.

Mid-Mesoproterozoic (1.4–1.2 Ga) rift-related and anorogenic magmatism is widely distributed in various cratons within the Columbia supercontinent (e.g., Zhao et al., 2004; Ernst et al., 2008 and references therein; Hou et al., 2008; Goldberg, 2010). One of the most important Mid-Mesoproterozoic large igneous province is located in the Canadian Shield of North America including the 1.27 Ga Mackenzie giant radiating mafic dyke swarms and coeval Copper River flood basalts and extends over an area of 2.71 million km^2 (Ernst and Buchan, 2001; Ernst et al., 2008). The mafic dyke swarms and coeval flood basalts were proposed to be related to a mantle plume at the northern apex of the radiating Mackenzie swarm (Ernst et al., 1995) during the final breakup of the Columbia supercontinent (Zhao et al., 2004; Hou et al., 2008). Similar mafic dyke (sill) swarms and related magmatic rocks are widely distributed in Greenland (Pearce and Leng, 1996; Cadman et al., 2001; Upton et al., 2003), Australia (Mortimer et al., 1988; Pidgeon and Nemchin, 2001; Pidgeon and Cook, 2003; Wingate et al., 2000, 2005), East Antarctica (Black et al., 1991; Lanyon et al., 1993; Hoek and Seitz, 1995), Baltica (Söderlund et al., 2005), Siberia (Ernst et al., 2000; Ernst and Buchan, 2001; Gladkochub et al., 2010), South America (Bettencourt et al., 1999; Ernst and Buchan, 2001; Pimentel et al., 2004), West Africa and Congo (Haggerty, 1982; Ernst and Buchan, 2001; Mayer et al., 2004), Kalahari (Hanson et al., 2006) and North China Cratons (Zhang et al., 2009b; Li et al., 2009a). These observations provide compelling indication that the 1.33–1.31 Ga magmatism in the northern NCC could be part of a major global magmatism in the Mid-Mesoproterozoic associated with the breakup of the Columbia supercontinent.

6.2. Tectonic setting of the Mid-Mesoproterozoic intrusive rocks

As shown in a previous section, the Mid-Mesoproterozoic intrusive rocks in the northern part of NCC are bimodal in nature as characterized by their chemical composition. The bimodal associa-

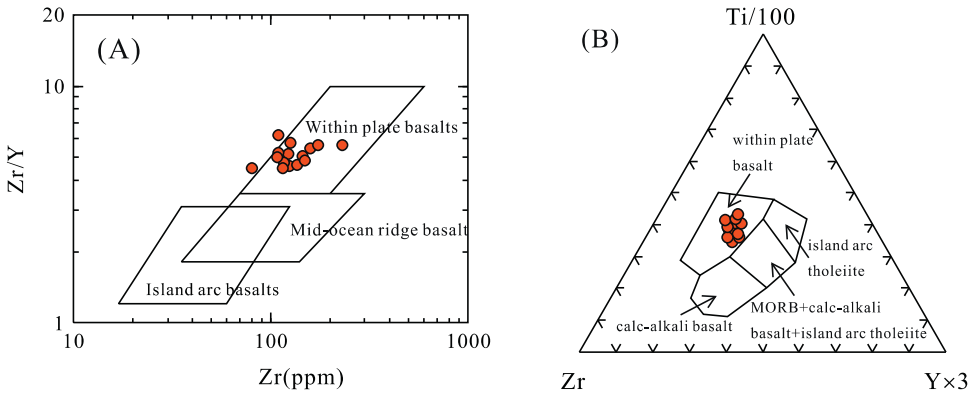


Fig. 14. Major and trace elements discrimination diagrams for Mid-Mesoproterozoic diabase sills from the northern part of the NCC. (A) Zr/Y vs. Zr diagram from Pearce and Norry (1979); (B) Ti–Zr–Y diagram from Pearce and Cann (1973).

tion suggests their genetic link with a continental rift environment (Wilson, 1989). Immobile trace elements of mafic rocks can provide important constraints on their tectonic environments (Rollinson, 1993). In the Zr/Y vs. Zr and Ti–Zr–Y discrimination diagrams (Pearce and Cann, 1973; Pearce and Norry, 1979), all the diabase samples fall within the fields of within plate basalts (Fig. 14), further indicating an intracontinental rift environment for these Mid-Mesoproterozoic intrusive rocks.

Most workers consider that there are at least two continental rift zones termed the Yanliao and Zha'ertai-Bayan Obo-Huade rifts in the northern NCC (Fig. 1) and that these rifts are probably related to the breakup of the Columbia supercontinent (Zhao et al., 2003a). However, the time of formation of these rift zones and the duration of rifting remain equivocal. Due to the lack of reliable isotopic ages for the Mesoproterozoic (1.6–1.0 Ga) magmatism and tectonism in the NCC, some of the previous models considered that continental rifting in the northern NCC culminated prior to 1.6 Ga (e.g., Rogers and Santosh, 2002; Zhai, 2004; Lu et al., 2008), represented by the 1.80–1.75 Ga mafic dyke swarms in the central part of the NCC and 1.78–1.75 Ga Xiong'er volcanic rocks (Peng et al., 2005, 2007, 2008). Recently, however, He et al. (2008, 2009, 2010a,b) and Zhao et al. (2009b) have shown that the Xiong'er volcanic rocks formed in an Andes-type continental margin arc. Moreover, most previous models suggested that the NCC was not involved in the final breakup of the Columbia supercontinent during 1.35–1.20 Ga (e.g., Rogers and Santosh, 2002; Hou et al., 2008). The evidence for Mid-Mesoproterozoic continental rift-related magmatism in the northern NCC as presented in this study is therefore crucial because it confirms that the NCC did experience Mid-Mesoproterozoic rifting events which led to the final breakup of the Columbia supercontinent.

6.3. Petrogenesis of the Mid-Mesoproterozoic bimodal intrusive rocks

The Mid-Mesoproterozoic diabase sills within the late Paleoproterozoic to Mesoproterozoic sedimentary rocks exhibit very similar petrological and geochemical compositions with low contents of SiO_2, high contents of TiO_2, Fe_2O_3T and MgO. They are classified as gabbro-gabbroic diorite or subalkaline basalt on total alkali ($K_2O + Na_2O$) vs. silica (SiO_2) and Nb/Y vs. Zr/TiO_2 diagrams (Fig. 10A–B) and exhibit tholeiitic compositions on the AFM diagram (Fig. 10C). These petrological and geochemical features indicate that the diabase rocks were generated from mantle-derived basaltic magma. Slight LREE-enrichment on chondrite-normalized REE patterns (Fig. 11A–E), positive Pb anomaly and slight negative Nb–Ta anomalies on primitive mantle-normalized spidergrams (Fig. 12A–E) suggest crustal assimilation during magma processes (Wilson, 1989). The whole-rock Nd and zircon and baddeleyite Lu–Hf isotopic data provide important constraints on the source characteristics. All the diabase samples exhibit very similar Nd and Hf isotopic compositions with positive whole-rock $\varepsilon_{Nd}(t)$ and zircon and baddeleyite $\varepsilon_{Hf}(t)$ values (Tables 4 and 5). On $\varepsilon_{Nd}(t)$ vs. emplacement age and $\varepsilon_{Hf}(t)$ vs. U–Pb age of zircons and baddeleyites, the rocks plot in the area between evolution lines of chondrite and depleted mantle (Fig. 13). These Nd and Hf isotopic systematics are very different from those of the 1.74–1.69 Ga Damiao anorthosite complex in the northern NCC that was considered to be derived from partial melting of an enriched lithospheric mantle (Xie, 2005; Zhang et al., 2007b). Partial melting of the subcontinental lithospheric mantle cannot account for the high positive $\varepsilon_{Nd}(t)$ and $\varepsilon_{Hf}(t)$ values of the diabase sills and the primary basaltic magmas of these rocks were likely derived from a depleted deep asthenosphere mantle source during the continental rifting stage. The Mid-Mesoproterozoic diabase sills in the northern NCC were likely generated by partial melting of the depleted asthenosphere mantle coupled with slight crustal assimilation in a continental rifting environment.

Rocks from the Mid-Mesoproterozoic granitic pluton are characterized by high contents of SiO_2, $K_2O + Na_2O$, low TiO_2, Fe_2O_3T and MgO and classified as granite or rhyodacite/dacite in geochemical classification diagrams. These felsic intrusive rocks can be produced either by fractional crystallization of a basaltic magma or by partial melting of the ancient continental crust. The Mid-Mesoproterozoic granitic rocks are characterized by low negative whole-rock $\varepsilon_{Nd}(t)$ and zircon $\varepsilon_{Hf}(t)$ values and old Nd and Hf isotopic model ages (Tables 4 and 5). Their Nd–Hf isotopic compositions are very different from those of the diabase sills, indicating different source areas for these mafic and felsic intrusive rocks. On $\varepsilon_{Nd}(t)$ vs. emplacement age and $\varepsilon_{Hf}(t)$ vs. U–Pb age of zircons, they fall into an area near the 2.4–2.6 Ga evolutionary trend line of ancient crust of the NCC (Fig. 13), indicating their derivation from ancient crustal material. Therefore, the Mid-Mesoproterozoic granitic rocks in the northern NCC were mainly produced by partial melting of the older continental crust. The crustal anatexis in the northern NCC in the Mid-Mesoproterozoic was likely induced by upwelling hot asthenosphere mantle during continent rifting processes.

6.4. Implications for reconstruction of the Columbia supercontinent

Although it is well recognized that the NCC was involved in the global assembly of the Columbia supercontinent during Paleoproterozoic time (e.g., Condie, 2002; Wilde et al., 2002; Zhao et al.,

2003a,b; Kusky et al., 2007; Santosh, 2010d), the position of this craton within the Columbia supercontinent is not well constrained because of the lack of reliable paleomagmatic and geochronological data. Models on the paleo-position of the NCC within the Columbia supercontinent in the Paleo-Mesoproterozoic are diverse, and proposals include the connection of the NCC to Siberia (Condie, 2002), Baltica (Wilde et al., 2002), India (Zhao et al., 2003b) or a juxtaposition between Baltica, South America and Africa (Kusky et al., 2007). Since the evidence for Meso-Neoproterozoic (1.6–0.5 Ga) magmatism and tectonism was lacking in the NCC, most researchers believe that the NCC was not involved in the final breakup of the Columbia supercontinent during 1.35–1.20 Ga (e.g., Rogers and Santosh, 2002; Hou et al., 2008) nor the assembly and breakup of the Rodinia supercontinent in the late Mesoproterozoic to Neoproterozoic (e.g., Lu et al., 2008). Thus, many paleogeographic reconstructions for the Columbia and Rodinia supercontinents during Meso-Neoproterozoic time have excluded the NCC (e.g., Hoffman, 1991; Dalziel, 1991; Rogers and Santosh, 2002; Pesonen et al., 2003).

The major Mid-Mesoproterozoic continental rift magmatism in the northern NCC documented in our study therefore provides important constraints for the reconstruction of the Columbia supercontinent and the timing of breakup of the NCC from the Columbia supercontinent. A comparison of the paleomagnetic data from the NCC with those of the Laurentia and Siberia (Zhang et al., 2009b), combined with the distribution of the Mid-Mesoproterozoic (1.4–1.2 Ga) dyke (sill) swarms, and the 2.1–1.8 Ga orogens indicates that the NCC was a member of the Nena amalgam including Laurentia, Siberia and Baltica cratons in the Columbia supercontinent. The NCC was probably located between the North America (Laurentia) and Siberia cratons in the late Paleoproterozoic to Mid-Mesoproterozoic (Zhang et al., 2009b). The breakup of the NCC from the Columbia supercontinent is recorded by the 1.33–1.31 Ga bimodal magmatism in the northern NCC and supported by paleomagmatic data (Halls et al., 2000; Wu et al., 2005; Pei et al., 2006; Zhang et al., 2006a; Li et al., 2008). Connections of the NCC with the North America (Laurentia) and Siberia cratons are also supported by similar late Paleoproterozoic to Mid-Mesoproterozoic sedimentary and volcanic successions in these cratons (e.g., Lu et al., 2002; Davidson, 2008; Pisarevsky et al., 2008; MacLean et al., 2009; Wang, 2011). Future studies should focus on paleomagmatic investigations on these large volumes of 1.33–1.32 Ga diabase sills in the northern NCC which would likely offer important constraints on the position of the NCC within the Columbia supercontinent.

6.5. Implications for genesis of the giant Bayan Obo REE–Nb–Fe ore deposit

The giant Bayan Obo REE–Nb–Fe ore deposit in northern NCC is among the world's largest known rare earth ore reserves and has attracted considerable attention since its discovery (e.g., Zhou et al., 1980; Drew et al., 1990; Chao et al., 1992; Yuan et al., 1992; Le Bas et al., 1992, 2007; Wang et al., 1994; Cao et al., 1994; Bai et al., 1996; Zhang et al., 2001; Liu et al., 2004; Smith, 2007; Wu, 2008). However, mineralization ages, tectonic setting and genesis of the deposits have been the subject of much debate for many years (e.g., Ren et al., 1994; Liu et al., 2004; Wu, 2008; Zhang et al., 2008). Although the ages of mineralization obtained from the REE–Nb deposit range from the Mesoproterozoic to Early Paleozoic, many results show that the Mid-Mesoproterozoic (1.4–1.2 Ga) is a very important period for REE–Nb mineralization (e.g., Nakai et al., 1989; Conrad and Mckee, 1992; Wang et al., 1994; Zhang et al., 1994, 2001; Liu et al., 2004, 2005). Both field relations and geochemical evidence show a close petrogenetic link between the carbonatites and the Bayan Obo REE–Nb mineral deposit (e.g., Bai and Yuan, 1985; Le Bas et al., 1992, 2007; Yuan et al., 1992; Wang et al., 2002; Yang et al., 2003, 2009). Although the carbonatite rocks in the Bayan Obo REE–Nb–Fe ore deposit have not be precisely dated, their Sm–Nd isochron and zircon U–Pb ages are mainly around 1.4–1.2 Ga (e.g., Zhang et al., 2001; Fan et al., 2006; Le Bas, 2006; Le Bas et al., 2007; Yang et al., 2010), indicating their emplacement in the Mid-Mesoproterozoic. Therefore, emplacement of the carbonatite rocks and REE–Nb mineralization in the Bayan Obo deposit were probably coeval with those of the diabase and granite rocks in the Yanliao and Zha'ertai-Bayan Obo-Huade rift zones in the northern NCC.

Global giant or large REE deposits are usually related to carbonatite complexes in continental rift zones and their geochemical features show that sources for the large-scale REE mineralization is closely related to input from anomalous subcontinental mantle during rifting processes (e.g., Lehmann et al., 1994; Bai et al., 1996; Niu et al., 2003; Xu et al., 2004; Huang et al., 2006; Ernst and Bell, 2010). Although many workers proposed the formation of the Bayan Obo REE–Nb–Fe ore deposit in a continental rift environment (e.g., Drew et al., 1990; Cao et al., 1994; Bai et al., 1996), whether the continental rift was still active in the northern NCC in the Mid-Mesoproterozoic (1.4–1.2 Ga) remains uncertain, mainly due to lack of reliable geochronological data on contemporaneous rift-related magmatism. The Mid-Mesoproterozoic bimodal magmatism in continental rift setting within the northern NCC reported in this study provides critical constraints for the formation of the giant Bayan Obo REE–Nb–Fe ore deposit in a continental rift environment related to the breakup of the Columbia supercontinent in the Mid-Mesoproterozoic.

7. Conclusions

(1) New zircon and baddeleyite U–Pb geochronological results show that emplacement of the Mid-Mesoproterozoic magmatic rocks (diabase sills and granitoids) in the northern NCC occurred in the Mid-Mesoproterozoic at around 1.33–1.31 Ga.
(2) The Mid-Mesoproterozoic magmatic rocks in the northern NCC constitute a typical bimodal magmatic association. The diabase sills were likely generated by partial melting of the depleted asthenosphere mantle coupled with slight crustal assimilation and the granitic rocks were mainly produced by partial melting of the ancient continental crust. These magmatic rocks are related to continental rifting in northern NCC in the Mid-Mesoproterozoic.
(3) The northern NCC experienced Mid-Mesoproterozoic rifting events that are considered to have led to the final breakup of the Columbia supercontinent. Breakup of the NCC from the Columbia supercontinent occurred in the Mid-Mesoproterozoic, probably at around 1.33 Ga.

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